INTERNATIONAL MONETARY FUND

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WP/26/228

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**2026**
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**IMF Working Paper**
African and Statistics Departments

# Macroeconomic Effects of Forced Displacement in Sub-Saharan Africa

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Authorized for distribution by Kangni Roland Kpodar
October 2026

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**ABSTRACT:** This paper estimates the macroeconomic effects of forced displacement on host economies in Sub-Saharan Africa, drawing on cross-country panel data and country case studies of Burkina Faso, South Sudan, and Uganda to provide granular evidence on the consequences of spatial concentrations of forcibly displaced persons within countries. Estimates from a difference-in-differences local projections (LP-DiD) framework do not suggest that large refugee inflows significantly affect host-country GDP, inflation, or the current account balance over a five-year horizon. The effects on the primary fiscal balance also appear to be muted, yet this masks offsetting fiscal dynamics: government expenditures rise and fiscal revenues increase (mainly through taxes). These results are robust to restricting the control group to Sub-Saharan African economies. Under a broader treatment definition that captures the total displaced population (refugees and internally displaced persons), the GDP response turns significantly negative and intensifies with displacement severity, though this estimate is best interpreted as an upper bound because it cannot be fully separated from the effects of conflict itself that generates displacement. The cross-country findings are consistent with the case studies, which indicate generally muted effects on aggregate food inflation and wages. Beneath this muted aggregate response, however, a consistent pattern emerges across cases: prices of imported and processed staples rise, while those of locally produced cereals stay flat or fall. The case studies also yield mixed evidence regarding the impacts of displacement inflows on consumption and labor force participation, with such inflows reducing host employment in Burkina Faso but raising it in Uganda.



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|---|---|
| JEL Classification Numbers: | F22, O15, J61, J68 |
| Keywords: | Forced Displacement; Fragile and conflict-affected states; Refugees; Sub-Saharan Africa |
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WORKING PAPERS

# Macroeconomic Effects of Forced Displacement in Sub-Saharan Africa

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# Macroeconomic Effects of Forced Displacement in Sub-Saharan Africa*

Hany Abdel-Latif[a], Antonio David[a,*], Mouhamadou Moustapha Ly[a], Elisee Wendlassida Miningou[a], Masateru Okamoto[a], Alex Vaval Pierre-Charles[a]

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## 1. Introduction

The global forced displacement crisis nears record levels, with around 118 million forcibly displaced people (FDP) at end-2025.[1] Sub-Saharan Africa (SSA) has experienced a particularly sharp increase in FDPs, rising from 8.7 million in 2010 to 40.9 million in 2025, about 35 percent of the global population of forcibly displaced. This surge, driven primarily by conflict and environmental disasters, poses substantial challenges for host countries. Quantifying the macroeconomic implications of forced displacement in SSA is therefore essential, given the region's limited resources and rising fiscal pressures, as well as the need for effective policy responses that preserve stability and support the welfare of both displaced populations and host communities.

This paper estimates the macroeconomic effects of forced displacement on host economies in SSA using both cross-country panel data and country-specific case studies for Burkina Faso, South Sudan, and Uganda. The case studies provide additional granularity by linking outcomes to the geographic concentration of FDPs within countries. To our knowledge, this is the first study to use cross-country panel data to systematically analyze forced displacement in SSA, while explicitly seeking to identify causal effects using the local projections difference-in-differences (LP-DiD) approach proposed by (Dube et al., 2025). Existing empirical work has largely emphasized micro-level, country-specific quasi-natural experiments based on household or labor market data, with comparatively limited attention to aggregate macroeconomic outcomes. By contrast, our cross-country perspective facilitates more generalizable inferences.

Large inflows of refugees and internally displaced people can affect macroeconomic outcomes through several channels (Table 1 provides a summary). For host economies, the most immediate mechanism is typically an expansion of labor supply, which can, in principle, support economic growth. However, misallocation may arise, with adverse implications for total factor productivity, if displaced people cannot

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The authors would like to thank Sneha Agrawal, Aliona Cebotari, Christian Ebeke, Nombulelo Gumata, Roland Kpodar, Frederic Lambert, Nils Lehr, Montfort Mlachila, Haonan Qu, Andrea Richter Hume, Stephane Roudet, Chima Simpson-Bell, and participants at the IMF's African Department seminar series for excellent comments and suggestions. The usual caveats apply.

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[1] This figure includes Other people of concern by the United Nations High Commissioner on refugees and Palestinian refugees under UNRWA, whom we exclude for the rest of the descriptive statistics and paper.

access jobs commensurate with their skills, or if institutional and other constraints hinder labor-market absorption. Refugee inflows may also raise domestic demand (especially consumption), thereby stimulating activity in the short run while exerting pressure on prices. At the same time, if such inflows increase perceived risks of social tensions and broader uncertainty, they may dampen economic activity, notably through weaker investment. On the fiscal side, large displacement episodes are likely to require additional near-term spending to provide essential services (health, education, housing, and humanitarian assistance), potentially deteriorating fiscal balances or inducing expenditure re-prioritization. These pressures may crowd-out development spending and public investment, with negative implications for long-run growth. Over longer horizons, however, displacement could strengthen fiscal revenues if displaced people are successfully integrated and the tax base expands. Regarding the external sector, large inflows would typically be expected to widen the current account deficit in the short term, consistent with higher fiscal outlays and private consumption. These pressures may be partly offset if inflows of displaced people are accompanied by additional international aid, remittances, or other transfers. Inflationary pressures may also affect the real exchange rate and external competitiveness, although moderation in real wage growth could mitigate these effects.

Finally, effects on labor market outcomes (wages, labor force participation, and employment) depend on the demographic and skills composition of displaced populations, as well as labor-market characteristics and rigidities in host countries. In the short run, a rise in informality is likely, and the expansion in labor supply may put downward pressure on wages.

The economics of forced displacement literature has typically focused on labor market outcomes and measures of well-being rather than broader macroeconomic effects. Verme and Schuettler (2021) conduct a meta-analysis of 59 studies on the economic effects of forced displacement on host communities, which largely comprise country-specific quasi-natural experiments using household surveys and other micro data, focusing on labor market outcomes, well-being, and prices. They conclude that effects on employment and wages are most often statistically insignificant. Food and rent prices tend to increase in the short term following large FDP inflows, whereas other prices may decline. As they note, an important limitation of this micro-based literature is that confounding factors, such as increases in central government spending or international aid associated with larger FDP inflows, are not appropriately captured by fixed effects.

Evidence for SSA is similarly concentrated on a limited set of events in a small number of host countries, primarily Tanzania and Sudan, with secondary attention to Uganda, Kenya, and Rwanda. For Tanzania, Ruiz and Vargas-Silva (2015) report a higher likelihood of self-employment or public-sector employment, alongside potential adverse effects on female employment outside the household. With respect to welfare and consumption, several studies find positive and statistically significant effects in Tanzania (Maystadt

and Duranton, 2019; Maystadt and Verwimp, 2014), as well as in Kenya (Alix-Garcia et al., 2018), Uganda (Kreibaum, 2016), and Rwanda (Taylor et al., 2016). However, evidence on food prices is more nuanced and appears to vary by product category. Exploiting the 1993–94 Burundian and Rwandan inflows into western Tanzania, Alix-Garcia and Saah (2010) find large price increases for non-aid staples such as plantains, bananas, and milk in markets close to the camps, whereas prices of goods included in the food-aid basket rose less or even declined, as imported food aid offset the additional demand. These price effects dissipate rapidly with distance from the camps, and translate into heterogeneous welfare effects: rural households, typically net sellers of agricultural goods, experienced gains in asset ownership, while urban households, typically net buyers, did not.

From a macroeconomic perspective, Alvarez et al. (2022) study the effects of Venezuelan migration flows on Latin American host countries using growth accounting and simulations based on the IMF's Flexible System of Global Models (FSGM), a multi-country semi-structural general equilibrium model. They find that higher demand and a larger labor force may have increased annual GDP growth in the largest recipient economies by 0.10–0.25 percentage points on average. They also estimate that migrant-related increases in public spending amount to 0.1–0.5 percent of GDP, depending on the country, while concluding that associated fiscal deficits should narrow over time as the tax base expands. Aiyar et al. (2016) examine the likely macroeconomic effects of the surge in asylum seekers in the European Union in 2015 on host countries also using a multi-country general equilibrium model. They conclude that in the short term, the refugee surge would lead to a very modest increase in GDP growth (less than 0.1 percentage points), reflecting the fiscal expansion associated with the costs of supporting the refugee population as well as an increase in labor supply. In turn, the medium and long-term growth effects would depend on how they will be integrated in the labor market.

Moreover, Bassanetti et al. (2025) use panel vector autoregressions for a group of countries in the Middle East and Central Asia to quantify the macroeconomic effects of refugee inflows. They find that a one percent increase in the refugee-to-population ratio is associated with a persistent rise in unemployment of about 0.2 percentage points, wider fiscal and current account deficits of roughly 0.1 percentage points over three years, and higher inflation of around 0.25 percentage points, while effects on growth are statistically insignificant. Countries with more conducive policies and legal frameworks for refugee inclusion appear to perform better following an inflow shock. Ebeke et al. (2026) evaluates the long-run fiscal effects of refugee inflows in SSA, and the extent to which inclusive refugee policies dampen these effects. Using a panel cointegration framework, the paper finds that refugee inflows tend to worsen the primary fiscal balance, but that inclusive policies mitigate this adverse impact.

Our estimates using the LP-DiD framework suggest that large refugee inflows do not have statistically

significant effects on GDP, inflation, or the current account balance over a five-year horizon. The effects on the primary fiscal balance are, in general, also muted, yet this masks offsetting fiscal dynamics: government expenditures rise, but so do fiscal revenues increase (mainly through taxes). When the treatment is broadened to the total displaced population (refugees plus internally displaced persons), the GDP response becomes significantly negative and intensifies with displacement severity; we interpret this cautiously, however, as an upper bound that cannot be cleanly separated from the effects of conflict driving displacement.

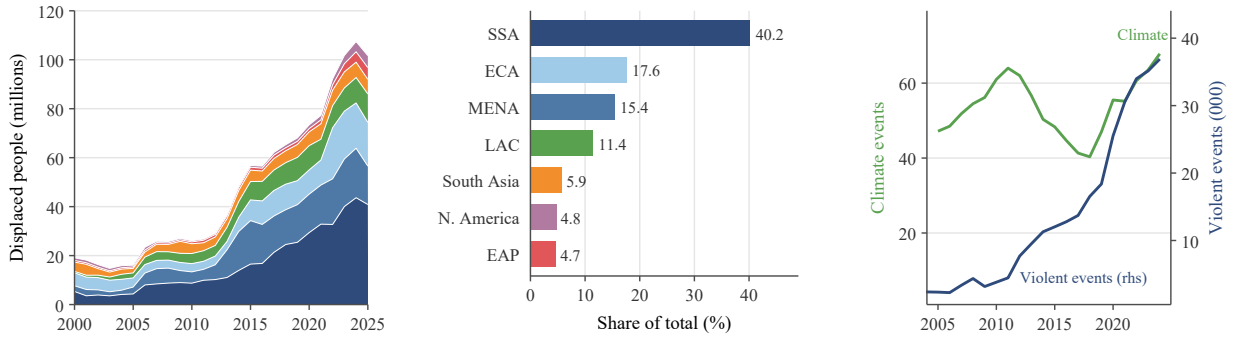
These results are robust to restricting the control group to SSA economies and are consistent with the case-study evidence. In particular, LP-DiD estimates for South Sudan indicate that effects on food inflation are statistically insignificant at shorter horizons but turn significantly negative at a twelve-month horizon; this aggregate decline reflects a sharp divergence across products, with imported and processed staples rising substantially while locally produced cereals stay flat or fall. Difference-in-differences estimates based on household surveys for Burkina Faso and Uganda similarly suggest that large inflows of forcibly displaced people do not materially affect aggregate food inflation in host communities, although they exert upward pressures on the prices of manufactured food products in Uganda. In addition, refugee inflows are associated with significant declines in household consumption, particularly among households engaged in agriculture and services. The case studies also point to mixed effects on labor force participation, reduced host employment in Burkina Faso and increased host employment in Uganda.

The remainder of the paper is organized as follows. Section 2 presents stylized facts on forced displacement in SSA. Section 3 describes the econometric methodology and data. Section 4 discusses results from the macroeconomic LP-DiD analysis and from the three household-survey-based case studies. Section 5 concludes.

## 2. Forced Displacement in Sub-Saharan Africa

Forced displacement has increased markedly since 2011 (Figure 1). While this upward trend is evident globally (Figure 1a), it has been especially pronounced in SSA, which accounts for more than one-third of the world's forcibly displaced population (Figure 1b). Figure 1c suggests that rising conflict intensity and the growing frequency of natural disasters have been key drivers of displacement across the region.

Figure 1: Forced Displacement, Climate and Conflict Events



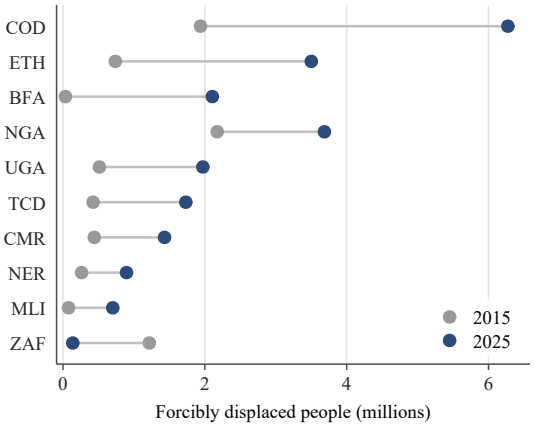

(a) Global forced displacement, 2000–25 (b) Share of forced displacement by region, 2025 (c) Climate and conflict events in SSA, 2005–24

*Note:* Colors in panel (a) correspond to regions in panel (b). SSA = Sub-Saharan Africa; MENA = Middle East & North Africa; ECA = Europe & Central Asia; LAC = Latin America & Caribbean; EAP = East Asia & Pacific. Panel (a) and (b) do not include other people of concern and Palestinian refugees under UNRWA.

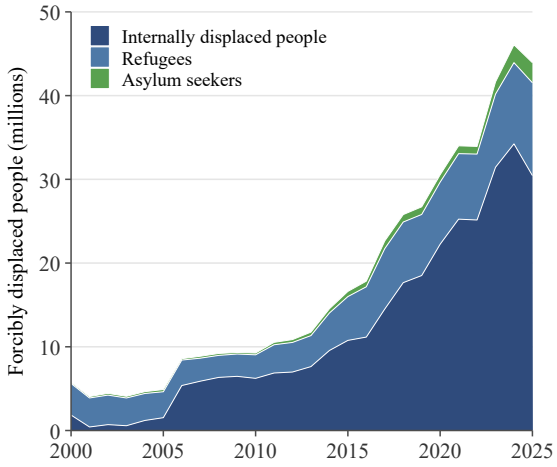
*Sources:* UNHCR Refugee Data Finder, Armed Conflict Location and Event Dataset Raleigh et al. (2010), EM-DAT, and IMF staff calculations.

Over the past decade, the Democratic Republic of Congo (DRC), Nigeria and Ethiopia have been among the most affected countries, each recording increases of more than 3 million forcibly displaced people with some countries like Burkina Faso and Uganda having close to 2 million forcibly displaced people, underscoring the scale and persistence of displacement in the region (Figure 2a). Unlike regions where cross-border movements dominate, forced displacement in SSA is characterized by a high incidence of internal displacement (Figure 2b). In 2025, the region recorded about 30 million internally displaced persons (IDPs), nearly three times the number of refugees. Recent data indicate that Uganda is the largest refugee-hosting country in the region in absolute terms, with close to 2 million refugees as of end-2025, followed by Chad and Ethiopia (Figure 2c). Uganda's relatively inclusive refugee framework, including access to land, education, and employment, has positioned it as a central actor in the regional response, notwithstanding its own development constraints. In terms of internal displacement, Sudan, the DRC, Nigeria, Somalia, Ethiopia, and Burkina Faso record the largest numbers of IDPs. In per-capita terms, Chad is the most affected country as of 2025 (with refugees accounting for roughly 7 percent of the population), followed by Mauritania, South Sudan and Uganda, while IDPs are particularly prevalent in Sudan, Somalia, South Sudan, Burkina Faso and the Central African Republic (Figure 2d). Displacement in the region is largely intra-regional as most refugees seek protection in neighboring countries, about 92.5 percent of refugees and asylum seekers hosted in SSA originate from a country that shares a border with their host. Figure 3 shows the flows of refugees by origin and host countries. It shows that in 2025 refugees hosted in Uganda primarily originate from South Sudan and the DRC. In Chad, they mainly originate from Sudan. In Ethiopia, they largely come from South Sudan, Somalia and to a lesser extent from Eritrea.

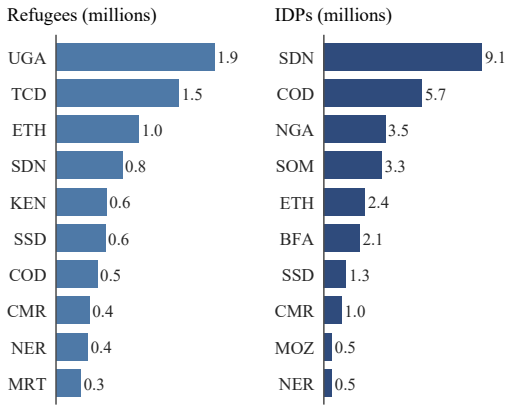
Figure 2: Forced Displacement Flows



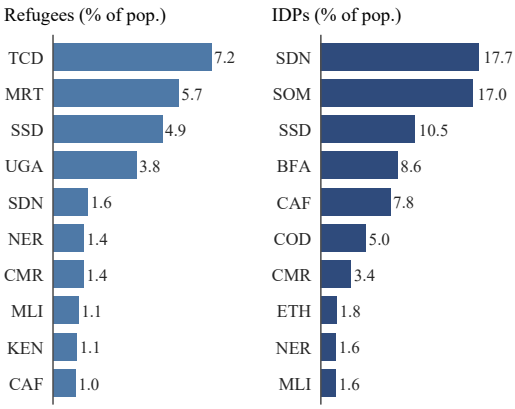

(a) Change in forced displacement in SSA, 2015–25




(b) Types of FD in SSA, 2000–25

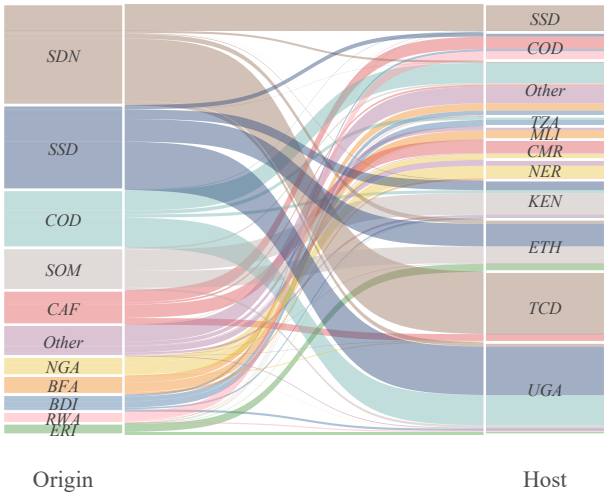



(c) Top Host Countries by Type of FD, 2025




(d) Top Host Countries by Type of FD, 2025

Figure 3: Flow of refugees in SSA (2025)




## 3. Data and Methodology

### *3.1. Methodology*

#### *3.1.1. Cross-country LP-DiD*

This paper employs quasi-natural experimental approaches to identify the effects of large forced-displacement flows on economic outcomes in host countries. At the cross-country level, the difference-in-differences (DiD) design compares countries that receive large numbers of refugees with those that do not. The DiD estimator captures the differential effect of the displacement shock on treated countries relative to the control group.

More specifically, we use panel data and local projections to estimate impulse responses that trace the dynamic effects of large refugee inflows across horizons. We adopt the LP-DiD specification developed by (Dube et al., 2025):

$$\Delta^h y_{i,t} \equiv y_{i,t+h} - y_{i,t-1} = \beta_h D_{i,t} + \sum_{l=1}^{2} \Gamma'_{h,l} X_{i,t-l} + \delta_t^h + \alpha_i^h + \varepsilon_{i,t}^h, \quad h = 0, 1, \ldots, H \tag{1}$$

where $y_{i,t}$ denotes the outcome of interest (e.g. log real GDP, log CPI, the current account balance-to-GDP ratio, and the primary fiscal balance-to-GDP ratio) for country $i$ in year $t$. $D_{i,t}$ indicates treatment status, a net refugee-inflow surge in country $i$ at time $t$, as defined in Section 3.2.1. $\beta_h$ is the coefficient of interest and captures the cumulative dynamic average treatment effect at horizon $h$.

$X_{i,t-l}$ is a vector of pre-treatment lagged covariates, including two lags of the differenced outcome variable. $\delta_t^h$ and $\alpha_i^h$ denote horizon-specific time and country fixed effects, respectively, absorbing common shocks

and unit-invariant characteristics at horizon $h$. $\varepsilon_{i,t}^{h}$ is the error term.

*3.1.2. Sub-national LP-DiD (South Sudan)*

For the South Sudan case study, we apply the LP-DiD specification in Equation 1 at the county-month level. Treatment $D_{i,t}$ now indicates that county $i$'s cumulative refugee-plus-returnee inflow share first exceeded the displacement threshold (see Section 3.2.2); the LP-DiD includes county and year-month fixed effects, and standard errors clustered at the county level. The high frequency of the monthly food-price panel makes the local-projections framework viable, in contrast to the household-survey panels for Burkina Faso and Uganda.

*3.1.3. Two-wave DiD (Burkina Faso and Uganda)*

For the Burkina Faso and Uganda case studies, the limited time dimension of the household-survey data does not support the local-projections framework. We therefore use a difference-in-differences (DiD) design that compares changes in outcomes between municipalities (or districts) hosting large numbers of forcibly displaced persons (treatment) and those with no or minimal inflows (control), before and after the shock. Exploiting temporal and spatial variation in inflows, the DiD setup helps isolate the effect of displacement from other confounding factors. The DiD specification is outlined in Equation 2:

$$y_{i,t} = \beta_0 + \beta_1 D_i + \beta_2 \text{Shock}_t + \beta_3 (D_i \times \text{Shock}_t) + \phi' X_{i,t} + \varepsilon_{i,t} \tag{2}$$

where $y_{i,t}$ denotes the outcome of interest (wages, consumption, inflation, or labor force participation); $D_i$ is an indicator equal to one for units affected by forced displacement (treatment) and zero for units with limited exposure (control); $\text{Shock}_t$ is an indicator equal to one in the post-event period and zero in the pre-event period; and $D_i \times \text{Shock}_t$ is the interaction term, which equals one only for treated units in the post period. $X_{i,t}$ denotes a set of control variables, and $\varepsilon_{i,t}$ is the error term. $D_i$ is the time-invariant analogue of the dynamic treatment indicator $D_{i,t}$ in Equation 1; the interaction $D_i \times \text{Shock}_t$ accordingly identifies the average treatment effect on the treated in the post-event period.

For the Burkina Faso case study, we additionally apply propensity score matching (PSM) to improve comparability between treated and control units on observable characteristics. Propensity scores are estimated using a logistic model with covariates including gender, urban/rural status, and region of residence; the estimated scores are used to construct weights applied in the DiD regressions reported in Section 4.2.1. For Uganda, treatment assignment is only weakly correlated with observable host-community characteristics, so PSM is unlikely to improve control-group comparability while substantially reducing the effective sample size available for estimation. For this reason, PSM is not implemented in the Uganda case

study. [2] An event-study analysis for Uganda finds no evidence of differential pre-treatment trends. A comparable test could not be conducted for Burkina Faso because only one pre-treatment period is available in the data. [3]

*3.1.4. Estimator and matching choices across cases*

Table 2 summarizes the four design choices that differ across the cross-country macroeconomic analysis and the three case studies. The choice of estimator is driven by the time dimension of the available data. The cross-country panel (annual, 1970-2023) and the South Sudan RTFP series (monthly, from January 2023 onward) both support the LP-DiD framework, while the Burkina Faso and Uganda household surveys are restricted to two analyzable waves and require the simpler two-wave DiD design. The choice of treatment variable and threshold reflects the underlying displacement category: refugees scaled to host-country population for the cross-country and Uganda settings; IDPs scaled to host-municipality population for Burkina Faso; combined refugees plus returnees scaled to host-county population for South Sudan. Propensity score matching is applied in the Burkina Faso case study, where geographic and demographic covariates strongly predict exposure to displacement, but not in the other settings where assignment is either better balanced (Uganda) or operates at a level of aggregation (countries and counties) that makes individual-level matching inapplicable.

Under the standard identifying assumption that treated and control units would have followed parallel trends absent treatment, both designs yield an unbiased estimate of the causal effect of forced displacement. The strategy also assumes no anticipation effects, i.e., outcomes in treated communities do not respond prior to the influx of forcibly displaced persons.

*3.1.5. Robustness framework*

The robustness checks reported across the cross-country macroeconomic analysis and the three case studies fall into three categories, which we apply consistently throughout the paper.

*Identification robustness* addresses whether the parallel-trends assumption underlying the DiD design is credible. For the LP-DiD analyses (cross-country and South Sudan), this takes the form of pre-trend coefficient diagnostics and, for the South Sudan short-run estimates, credible DiD bounds in the framework of Rambachan and Roth (2023). In the case of the two-wave DiD designs in Burkina Faso and Uganda, we cannot test parallel trends directly; identification credibility there rests on the matching and threshold-restriction

[2] Applying PSM does not substantially change the results for Uganda, but the resulting reduction in sample size weakens the statistical power of the estimates, especially in subsample analyses. Although there may be concerns about the consistency of the methodology used across the Uganda and Burkina Faso case studies, we prefer the results without PSM, as they use a larger sample and appear more reliable.

[3] The results are available upon request.

choices summarised in Section 3.1.4.

*Estimator robustness* assesses whether the headline Average Treatment Effect on the Treated (ATT) survives alternative weighting and matching schemes. The Burkina Faso case study reports this dimension explicitly, comparing the baseline PSM specification with OLS, inverse-probability weighting (IPW), augmented IPW (AIPW), and the doubly robust Callaway-Sant'Anna estimator.

*Sample-restriction and treatment-definition robustness* assesses how sensitive results are to plausible variations in the control group or the treatment indicator. The cross-country analysis reports both an SSA-only sample restriction (Section 4.1.2) and a broader total-displacement treatment definition (Section 4.1.3). The Burkina Faso and South Sudan analyses examine spatial spillover by restricting the control group to non-adjacent units. The Uganda analysis compares a binary settlement-district treatment with a continuous refugee-share specification. Table 3 summarizes which test is conducted in which analysis.

### *3.2. Data and Sample Construction*

Data on refugees and their geographic distribution used in this paper are obtained from the United Nations High Commissioner on Refugees (UNHCR) portal(United Nations High Commissioner for Refugees (UNHCR), 2026), which reports the number of refugees and internally displaced persons, along with countries of origin, back to the 1960s. The portal also provides sub-national information on the distribution of forcibly displaced persons (FDPs), which we exploit for the case studies.

#### *3.2.1. Cross-country macroeconomic panel*

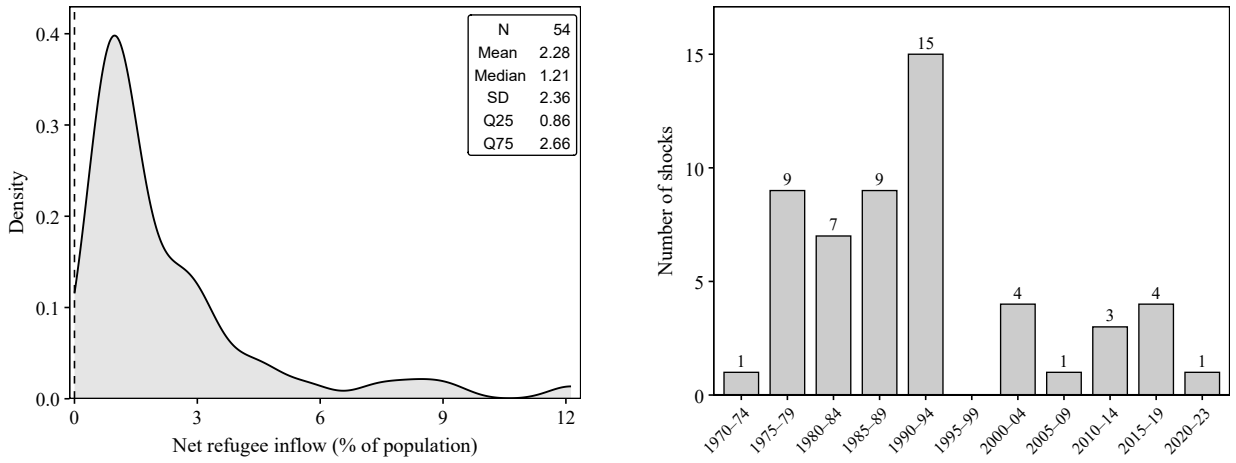
Macroeconomic data for the main outcomes (GDP, inflation, the current account balance-to-GDP ratio, and the primary fiscal balance-to-GDP ratio) come from the IMF's World Economic Outlook database (October 2025 vintage)(International Monetary Fund, 2025).

We construct an initial shock variable as follows: a country i is defined as experiencing a shock at time t when the net refugee inflow $I_{it}$, as a share of the population, exceeds 1.5 standard deviations above its country-specific historical mean. This criterion targets large inflow episodes from the host-country perspective. We also impose a minimum threshold of 0.5 percent of the population to ensure that the episode constitutes a sizeable shock.

$$S_{i,t} = \begin{cases} 1 & \text{if } \frac{I_{it}-\mu_i}{\sigma_i} > 1.5 \text{ and } I_{it} > 0.5 \\ 0 & \text{otherwise} \end{cases}$$

where $\mu_i$ and $\sigma_i$ are the country-specific long-run mean and standard deviation of refugee inflows, respectively. Figure 4 characterizes the 54 refugee-inflow shock episodes in SSA countries identified in our sample along two dimensions. The panel (a) shows the distribution of net refugee inflows (as a share of population) conditional on a shock occurring. The distribution is right-skewed, with a median inflow of 1.21 percent of the host-country population and a mean of 2.28 percent, reflecting that most shocks are relatively contained. Nonetheless, the sample includes very large shocks, such as Somalia in 1979–1980 and Djibouti in 1989–1990. The panel (b) plots the number of shock episodes by five-year period from 1970 to 2023. Shocks are concentrated in two distinct waves: a first wave spanning the late 1970s through the mid-1990s, peaking in 1990–1994 with 15 episodes, coinciding with major displacement crises (the Rwandan genocide, civil wars in Liberia and Sierra Leone, and the conflict in Mozambique); and a more modest second wave from 2010 onward, reflecting more recent displacement dynamics in East and Central Africa. Table 4 reports the shocks identified using our method.

Figure 4: Refugee Inflow Shocks in SSA




(a) Distribution of Net refugee inflows shocks

(b) Temporal pattern of refugee inflow shocks in SSA

*Note:* Sample restricted to SSA country-years. Net inflows are expressed as a percent of population.

A necessary condition for LP-DiD estimation is a valid comparison between treated and control units satisfying the parallel-trends assumption e.g. in the absence of treatment, outcome trajectories for treated and control units would have evolved similarly. While this assumption cannot be tested directly, examining pre-treatment differences can provide suggestive evidence of comparable trends prior to treatment. The principle that identification in staggered designs requires a clean comparison group, units untreated in the relevant period, whether never-treated or not-yet-treated, is established for the absorbing-treatment case by Callaway and Sant'Anna (2021) and Sun and Abraham (2021). Because our treatment is non-absorbing, a country may experience more than one inflow shock over the sample, we follow Dube et al. (2025), constructing clean treated and control groups, such that estimated dynamic responses are not contaminated by other, overlapping displacement events. We build these groups as follows.

Let $A_{i,t}$ denote the occurrence of any displacement event in country $i$ at $t$,

$$A_{i,t} = \max\{S_{i,t},\, O_{i,t},\, U_{i,t}\},$$

combining the inflow shock $S_{i,t}$ defined above with refugee-origin shocks $O_{i,t}$ and refugee-outflow shocks $U_{i,t}$. A country $i$ is cleanly treated at $t$, for horizon $h$, if it experiences an inflow shock, was free of any displacement event over the preceding $\ell$ years, is an SSA country and remains free of any such event over the following $h$ years:

$$D_{i,t}^{(h)} = \mathbf{1}\{i \in \text{SSA}\} \cdot S_{i,t} \cdot \mathbf{1}\left\{\sum_{s=1}^{\ell} A_{i,t-s} = 0\right\} \cdot \mathbf{1}\left\{\sum_{s=1}^{h} A_{i,t+s} = 0\right\},$$

where $\ell$ is the look-back window ($\ell = 5$ in our baseline) and $h$ the estimation horizon. We impose the symmetric requirement on controls: an admissible control at horizon $h$ has no inflow shock at $t$ and satisfies the same two-sided clean-window restriction but do not restrict the control be SSA countries,

$$C_{i,t}^{(h)} = (1 - S_{i,t})(1 - O_{i,t})(1 - U_{i,t}) \cdot \mathbf{1}\left\{\sum_{s=1}^{\ell} A_{i,t-s} = 0\right\} \cdot \mathbf{1}\left\{\sum_{s=1}^{h} A_{i,t+s} = 0\right\}.$$

The estimation sample at horizon $h$ then consists of the cleanly treated and cleanly untreated observations, $\{D_{i,t}^{(h)} = 1\} \cup \{C_{i,t}^{(h)} = 1\}$.

Two features distinguish this construction. First, whereas the standard clean-control criterion prunes only on recurrence of the treatment itself, we exclude any inflow, refugee origin, or refugee outflow shock. The three restrictions embodied in $A_{i,t}$ have concrete interpretations for a control observation. First, it must not have experienced a large refugee-inflow shock (the treatment). Second, it must not be a major country of refugee origin, generating an unusually large outflow of its own population seeking refuge abroad, since such episodes typically reflect conflict or political dynamics that would invalidate the unit as a counterfactual. Lastly, it must not have experienced a large, sudden return or departure of the refugee stock it hosts, which could generate distinct macroeconomic effects.[4] A host country that is simultaneously, or subsequently, a country of refugee origin or outflow is subject to displacement dynamics operating through channels other than the inflow we study, and retaining such observations would confound the estimated response with these co-occurring events. Second, we apply these conditions in both the pre- and post-treatment windows and to treated and control units alike. Most countries in SSA experience some form of forced displacement, so the number of units meeting these conditions in any given period is limited. To address this, we expand the control group beyond SSA, allowing other Emerging and Developing Economies to enter the control set

[4] In practice, this third case does not occur for any country in our sample.

provided they satisfy the same clean-control conditions. Our final treated group sample is composed of 33 shocks across 30 countries.

### *3.2.2. South Sudan county-month panel*

South Sudan offers an unusually clean setting in which to study the pass-through of forced displacement to consumer food prices. The April 2023 outbreak of war in Sudan triggered a sharp and geographically uneven inflow of South Sudanese returnees and Sudanese refugees: by December 2025 the cumulative total reached 1.26 million—about 10 percent of South Sudan's population—with returnees accounting for roughly two-thirds of the flow.[5] This shock is overlaid on a longstanding internal-displacement crisis from the 2013–2020 civil war, but its geographic concentration in border counties of the north creates substantial cross-county variation in exposure suitable for difference-in-differences identification.[6]

Monthly food prices come from the World Bank Real-Time Food Prices (RTFP) database, which covers 37 markets across 31 counties from January 2023 through November 2025.[7] A county is classified as treated in any month from April 2023 onward in which its cumulative inflow share—refugees plus returnees, scaled by 2024 population—first exceeds 8 percent, the average internal-displacement share observed before the August 2018 ceasefire of South Sudan's earlier civil war.[8] Returnees data are also corroborated by the International Organization for Migration (IOM).

### *3.2.3. Burkina Faso household panel*

Internal displacement in Burkina Faso has surged since 2019 amid escalating violence and insecurity, exceeding 1.5 million people by 2021 and 2 million by 2023. Displacement was initially concentrated in the north, but has since spread to other regions.

The analysis uses a two-wave panel household survey conducted in 2018 (pre-surge) and 2021 (post-surge), drawn from the World Bank's Living Standards Measurement Study (LSMS).[9] The survey tracks the same individuals across waves, controlling for time-invariant unobserved individual characteristics. Treatment is defined at the municipality level: a municipality is treated if IDPs represent at least 0.05 percent of its host

[5] UNHCR. `https://data.unhcr.org/en/situations/sudansituation` (last accessed December 23, 2025).

[6] The number of refugees from South Sudan's earlier civil war reached over 2.3 million in March 2023, including 808,336 in Sudan. See UNHCR `https://data.unhcr.org/en/documents/details/100546` (last accessed December 23, 2025).

[7] Market-level prices are averaged within each county-month. County-level population estimates are obtained from the United Nations Office for the Coordination of Humanitarian Affairs (UNOCHA), `https://data.humdata.org/dataset/cod-ps-ssd?` (last accessed December 23, 2025).

[8] In June 2018, 843,234 IDPs were monitored (IOM. `https://dtm.iom.int/south-sudan`, last accessed December 23, 2025). Monthly refugee and asylum-seeker presence at camps and transit centres is obtained from UNHCR `https://data.unhcr.org/en/dataviz/385` (last accessed December 23, 2025); monthly returnee inflows are constructed by aggregating IOM Displacement Tracking Matrix Event Tracking data; both are aggregated to the county level. Out of 79 counties (including the disputed Abyei area), 13 hosted refugees and asylum-seekers in 21 camps and transit centres as of November 2025.

[9] Although three LSMS rounds are available for Burkina Faso (2014/15, 2018/19, 2021/22), our DiD design exploits the pre- and post-2019 contrast that is best identified by the two 2018/19 and 2021/22 waves.

population, and serves as control otherwise. Although this threshold is necessarily somewhat arbitrary, it ensures that control municipalities have limited exposure to displacement-related shocks.

#### *3.2.4. Uganda household panel*

Uganda has experienced large refugee inflows in recent years, driven primarily by instability in South Sudan and the Democratic Republic of the Congo. These arrivals increased sharply from 2016 onward following the escalation of political violence in South Sudan, bringing the total number of refugees to nearly 1.5 million by 2020. Although the humanitarian dimensions of these movements are well documented, their economic consequences for host communities remain less well understood.

We apply the two-wave DiD framework (Section 3.1.3) to the LSMS panel survey, comparing outcomes prior to the 2016 surge (2011, 2012, and 2014 waves) with outcomes thereafter (2015/16, 2018/19, and 2021/22 waves). Treatment is defined at the district level: a district is treated if refugees account for at least 0.05 percent of the host population. These household data are matched with UNHCR refugee statistics.

## 4. Results

### *4.1. Macro Evidence of Forced Displacement Impacts*

#### *4.1.1. Baseline: The effects of refugee inflows*

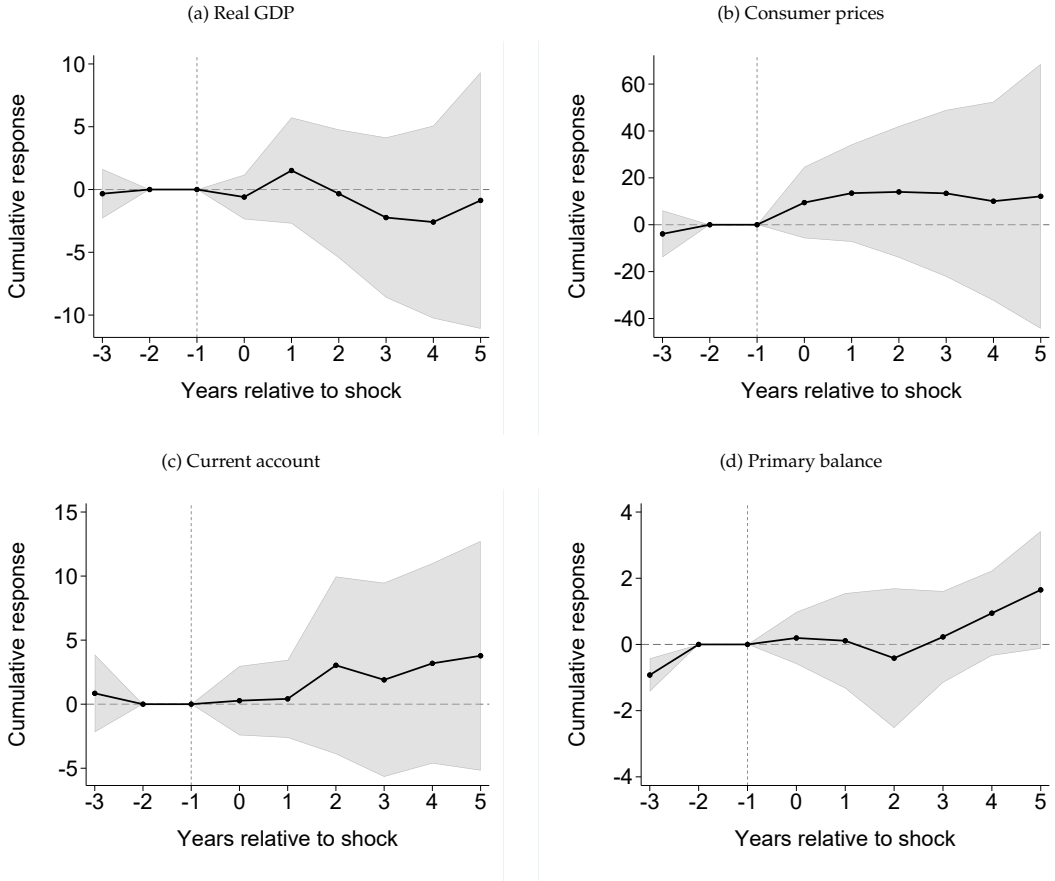
The impulse responses estimated using the LP-DiD approach over a five-year window around the shock are shown in Figure 5; the corresponding regression results are reported in the appendix. We discuss each outcome in turn.

Before interpreting the post-shock impulse responses, we examine the pre-trend coefficients as a diagnostic for the parallel-trends assumption underlying our LP-DiD design.[10] Because the estimation sample changes across horizons, these pre-shock coefficients should be interpreted as a validation exercise rather than as a comparison of a single, fixed pre-treatment trajectory. In the baseline specification, which conditions on the lagged first difference of the outcome, the coefficient at $h = -3$ is not statistically distinguishable from zero at the 10% level for real GDP, consumer prices or the current account; nevertheless for the primary balance it is significant, pointing to a modest differential movement in the fiscal position in the years before the shock. Since conditioning on $\Delta y_{i,t-1}$ mechanically absorbs the nearest pre-treatment horizon and leaves $h = -3$ as the only informative pre-shock estimate in the baseline, we complement this with a test of unconditional parallel trends in which the lagged outcome dynamics are omitted, so that no pre-treatment coefficient is zero by construction (Table 5). Over the five years preceding the shock ($h = -5$ to $h = -2$), the

[10] The estimate at $h = -1$ is zero by normalization, since $y_{i,t-1}$ is the base period of the dependent variable. The estimate at $h = -2$ is zero by construction: the dependent variable at that horizon, $y_{i,t-2} - y_{i,t-1}$, is the negative of the lagged first difference $\Delta y_{i,t-1}$ included as a control.

unconditional coefficients for real GDP, consumer prices and the current account are small and statistically indistinguishable from zero at every horizon. The primary balance again displays a significant coefficient at $h = -3$ (−1.2 percentage points of GDP), but the adjacent horizons are small and insignificant, and the pooled average over $h \in [-5, -2]$ is −0.54 percentage points and not statistically significant, similar to the other outcomes, suggesting no differential trend on average. The pattern is broadly similar when extending the placebo 10 year[11] before the shock with marginal violation of pre-trends for real GDP around $h = -10$ and $h = -9$ though no violation closer to the shock. Taken together, this suggests that treated and control countries followed broadly similar macroeconomic trajectories prior to the refugee shock, supporting a causal interpretation of the post-shock estimates.

Figure 5: Macroeconomic response to a refugee inflow shock




*Note:* The figure reports the estimated dynamic effects of refugee inflow shocks using the baseline LP-DiD specification in Equation (1) on the log of real GDP × 100, log of headline CPI × 100, Current Account (percent of GDP) and Primary Balance (percent of GDP). The solid lines show the point estimates and the shaded areas indicate 90 percent confidence intervals. Horizon $t - 1$ is set to 0.

The estimates indicate that large refugee inflows are not associated with statistically significant changes in real GDP over the five-year post-shock horizon. Point estimates are initially modestly positive at $h = 1$ then

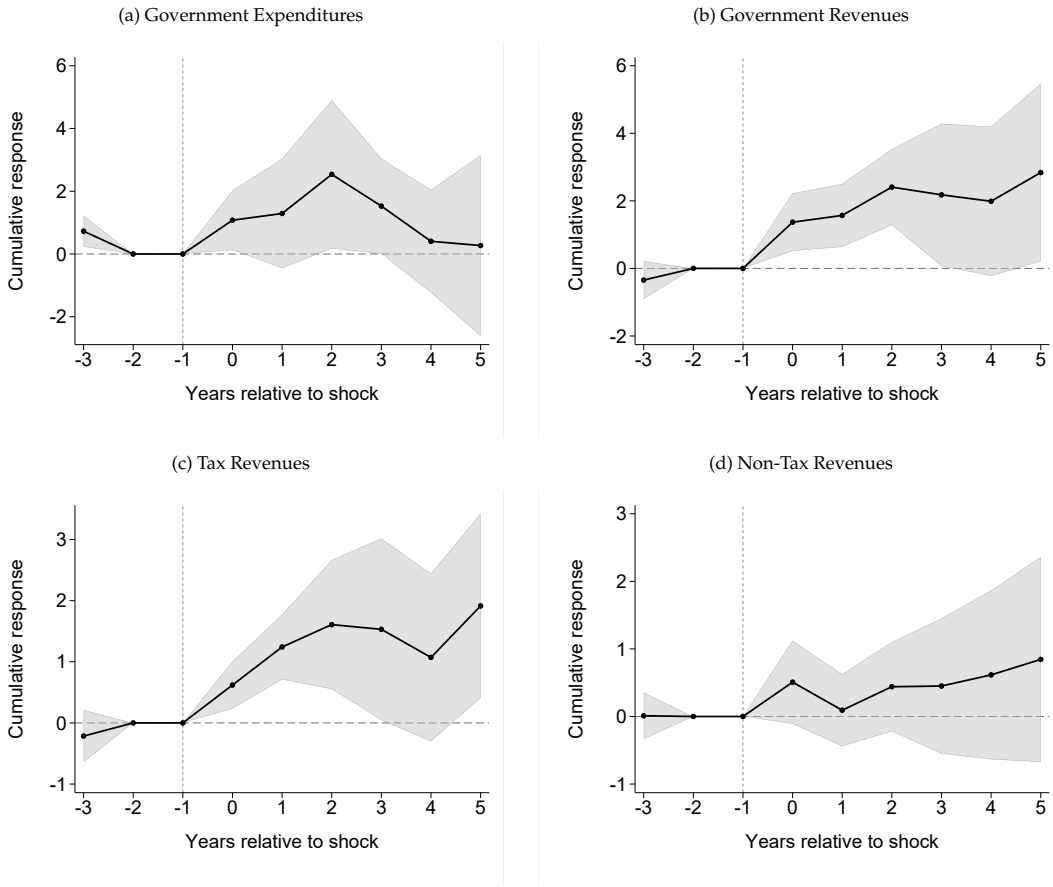
[11] We show placebo estimates for up to 10 periods before the shock though the sample varies substantially the further out the estimation horizon is.

turn modestly negative throughout but are not statistically significant at the 10 percent level. While CPI point estimates are positive and rise over time, they are statistically indistinguishable from zero, indicating muted inflation effects. The wide confidence intervals and lack of precise effects are consistent with the mixed evidence in the existing literature. The current account balance improves gradually, but the estimates are not statistically significant at any horizon.

The effects on the primary fiscal balance are also muted. The point estimate is essentially zero at $h = 0$ and $h = 1$, before turning marginally negative at $h = 2$ ($-0.42$ percentage points of GDP), again without statistical significance. The response then turns positive in later horizons, reaching 1 percentage point of GDP at $h = 4$ and 1.65 percentage points at $h = 5$. The primary balance result may mask the underlying fiscal adjustment. We could find no effect either because revenue and expenditure both respond little, or because both respond to the shock by similar amounts. To understand what lies behind this response, we estimate Equation (1) separately for government expenditures and revenues, further decomposing revenue into tax and non-tax components, all as a share of GDP. Figure 6 reports these results. Refugee inflow shocks raise both government expenditures and revenues, with the latter offsetting the former and presenting more persistent effects. Expenditures rise by about 1 percentage point of GDP at $h = 0$ and peak at 2.5 percentage points at $h = 2$. It then unwinds, falling back towards zero and becoming statistically indistinguishable from zero from $h = 4$ onward. The revenue response is clearer and more persistent. Total revenue increases by 1.37 percentage points of GDP on impact and by 1.57 and 2.41 percentage points at $h = 1$ and $h = 2$, each significant at the one percent level, and remains positive and significant at $h = 3$ and $h = 5$, reaching 2.84 percentage points by the end of the horizon.

Decomposing revenue into its tax and non-tax components shows that the aggregate response is driven by taxation. Tax revenue as a share of GDP rises by 0.62 percentage points on impact and by 1.24 percentage points at $h = 1$, both significant at the one percent level, reaching 1.61 percentage points at $h = 2$ and 1.91 percentage points by $h = 5$; the response is significant at every horizon except $h = 4$. Its path closely tracks that of total revenue in both sign and magnitude, indicating that taxation is the primary channel through which host governments partially offset the expenditure costs of refugee hosting. Non-tax revenue, by contrast, is positive at every horizon but never statistically distinguishable from zero, with point estimates ranging from 0.09 to 0.84 percentage points of GDP.

Figure 6: LP-DiD - Expenditures and Revenues




*Note:* The figure reports the estimated dynamic effects of refugee inflow shocks using the baseline LP-DiD specification in Equation (1) on the government expenditure (excluding interest payments), government revenue, tax revenue and non-tax revenue all as a share of GDP. The solid lines show the point estimates and the shaded areas indicate 90 percent confidence intervals. Horizon $t-1$ is set to 0.

The positive tax-revenue response is consistent with a demand-side fiscal channel operating through refugee-related consumption. Refugees and the humanitarian ecosystem surrounding them, including aid workers, NGO staff, and international contractors, increase local demand for goods and services subject to indirect taxes (e.g., value-added taxes and excise duties). This channel is particularly relevant in SSA, where indirect taxes (notably VAT and trade taxes) account for a large share of government revenue, implying a relatively direct pass-through from local consumption to government receipts.

Nevertheless, the fact that the revenue response persists after the expenditure response fades points beyond a purely cyclical demand channel. Absorption-driven effects would decay alongside the humanitarian operation itself. This persistence could be explained by a number of factors. For example, the initial public spending pressures following large refugee inflows could prompt governments to implement revenue mobilization measures (both in terms of tax policy and tax administration), which could have persistent effects on revenue collection. Taken together, these results indicate that a refugee shock places initial

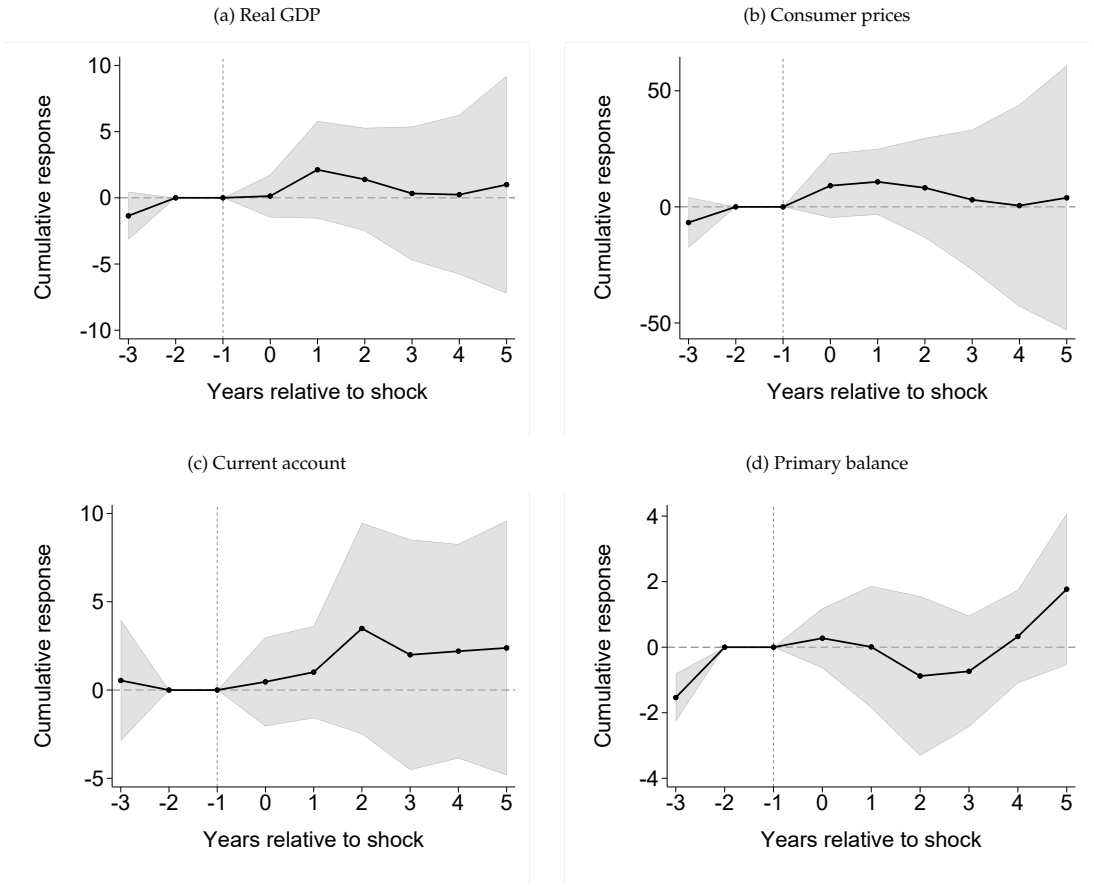
pressure on public finances as spending rises, but that this is matched by a comparable increase in revenue which persists after the expenditure response has faded.

#### *4.1.2. Robustness Checks*

*Sample-restriction robustness.* To assess the robustness of the results, we re-estimate the LP-DiD model in Equation 1 while restricting the control group to SSA countries only. This restriction likely increases the comparability of treated and control units, but at the cost of a smaller sample. Figure 7 reports the corresponding impulse responses.

The SSA-only estimates broadly replicate the baseline findings. The GDP response remains statistically insignificant at all horizons, with point estimates ranging from 0.13 percent at $h = 0$ to a peak of 2.1 at $h = 1$ and 1 by $h = 5$, the confidence bands are wide throughout. Point estimates for the effects on inflation are smaller than in the baseline and remain statistically insignificant. The effects on the current account are statistically indistinguishable from zero at every horizon, as in the baseline. The initial effects on the primary fiscal balance are negative reaching $-0.9$ percentage points at $h = 2$, before eventually turning positive and reaching 1.8 percentage points at $h = 5$, though the results are not statistically significant.

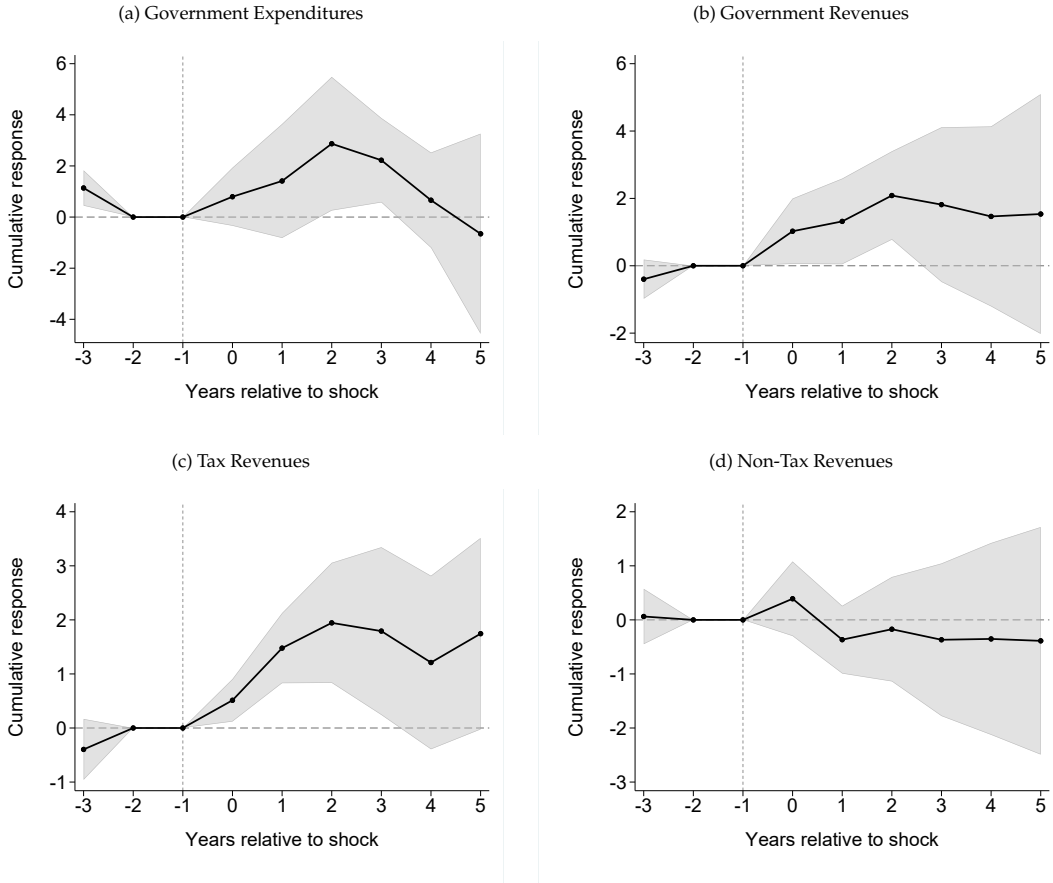
Figure 7: LP-DiD Estimates with Control Group Restricted to SSA Countries




*Note:* The figure reports the estimated dynamic effects of refugee inflow shocks using the baseline LP-DiD specification in Equation (1) on the log of real GDP × 100, log of headline CPI × 100, Current Account (% of GDP) and Primary Balance (% of GDP) restricting the sample to SSA control countries. The solid lines show the point estimates and the shaded areas indicate 90 percent confidence intervals. Horizon $t-1$ is set to 0.

Results for government expenditure and taxation are also broadly confirmed when restricting the sample to SSA economies. Government expenditure rises to 2.87 percentage points of GDP at $h = 2$ and 2.22 at $h = 3$, significant at the 10% and 5% levels respectively, before unwinding. This indicates that the expenditure channel is not driven by the inclusion of non-SSA control countries. Total revenue is positive and significant on impact and at $h = 1$ and $h = 2$, reaching 2.09 percentage points, but loses significance thereafter, so that the persistence observed in the baseline is less precisely estimated within the SSA sample. Tax revenue remains positive and significant at every horizon except $h = 4$, rising from 0.51 percentage points on impact to 1.95 at $h = 2$ and 1.75 by $h = 5$, closely tracking the baseline path. Non-tax revenue is negative from $h = 1$ onward and statistically insignificant throughout; the sign differs from the full-sample estimates, but in neither sample is the response distinguishable from zero. Pre-treatment coefficients at $h = -3$ are significant for government expenditure (1.14, $p < 0.01$) and, in the baseline outcomes, for the primary balance (−1.53, $p < 0.01$), so the SSA-only estimates for these two series should be read with the same caution as the corresponding baseline results.

Figure 8: LP-DiD Estimates with Control Group Restricted to SSA Countries




*Note:* The figure reports the estimated dynamic effects of refugee inflow shocks using the baseline LP-DiD specification in Equation (1) on the government expenditure (excluding interest payments), government revenue, tax revenue and non-tax revenue all as a share of GDP. The solid lines show the point estimates and the shaded areas indicate 90 percent confidence intervals. Horizon $t-1$ is set to 0.

*Controlling for conflict intensity and natural disasters.* An important identification concern is whether large refugee inflows are systematically directed toward countries with particularly favorable economic prospects, in which case the estimated effects of refugee shocks would partly reflect endogenous destination selection. Our strategy does not require the allocation of refugees across host countries to be literally random. It relies instead on two features of forced displacement. First, large displacement episodes are triggered by severe push factors in the country of origin such as conflict, persecution, or disaster that are plausibly exogenous to the host economy's own trajectory. Second, the initial destination choices of forced migrants are far more constrained than those of economic migrants: because the immediate objective is to reach safety, geographic proximity, accessibility, and the security situation of potential hosts are the primary determinants of where displaced populations first arrive. The empirical literature contrasts a view of refugees as forced migrants who seek protection in the nearest safe location with a view of them as opportunists who select destinations on the basis of prosperity, and finds strongly for the former. In a global study of bilateral flows, Moore and Shellman (2007) show that proximity and the presence of co-national networks dominate the choice

of asylum country. Hatton (2009) likewise finds that violence and persecution account for most of the variation in asylum applications, while destination-country policies exert only a secondary, deterrent effect. These patterns are reflected in the aggregate data: at the end of 2024, roughly two-thirds of the world's refugees were hosted in a country neighboring their country of origin (UNHCR, 2025). Much of this quantitative literature, however, examines European or other high-income destinations (Neumayer, 2005; Hatton, 2009; Di Iasio and Wahba, 2024). Systematic evidence on destination determinants in South–South, and specifically Sub-Saharan African, settings is comparatively scarce. Alongside global analyses such as Moore and Shellman (2007), the study by Lanati and Thiele (2024) is among the few to address these flows directly. Their findings echo the broader pattern that conditional on safety and proximity, refugees relocate primarily to neighboring countries and the constraints that underpin our identifying assumption are, if anything, likely to bind more tightly in Sub-Saharan Africa, where the feasible set of accessible and safe destinations for populations fleeing conflict is typically limited to contiguous states[12].

This does not imply that destination choice is entirely independent of host-country characteristics. Among destinations that are accessible and sufficiently safe, refugees may still sort according to economic conditions, public services, pre-existing migrant and refugee networks, and asylum or labor-market policies (Neumayer, 2005; Hatton, 2020). Consistent with this, Lanati and Thiele (2024) find that, conditional on safety and proximity, higher per-capita income and the availability of health and education services attract South–South refugee flows, and Di Iasio and Wahba (2024) identify pre-existing migrant and asylum-seeker networks as the single strongest pull factor shaping where asylum seekers locate. Two considerations nonetheless limit the implications of this margin for our design. First, the economically driven component of refugee mobility appears largely at a later stage, once safety has been secured, rather than in the initial cross-border flight. Studying the internal mobility of Syrian refugees within Turkey, Beine et al. (2021) show that refugees relocate in response to income differentials, yet remain more sensitive to distance and move over shorter distances than economic migrants, so that even this secondary sorting is comparatively constrained. Second, the destination attribute that would most directly threaten our identification, favorable contemporaneous macroeconomic prospects and absorptive capacity, receives at most a secondary, conditional weight in the initial destination decision, well behind proximity, safety, and pre-existing co-national networks; and networks, which the literature identifies as the strongest of the pull factors, reflect historical settlement rather than the current conditions that could confound our estimates (Moore and Shellman, 2007; Di Iasio and Wahba, 2024). As such, we argue that large refugee shocks to host countries are, at least initially, exogenous to current macroeconomic conditions, with economically driven sorting emerging only subsequently and at a finer, within-country scale. Forced displacement thus differs from standard economic migration not

[12] Using UNHCR data on refugee stocks by country of origin and country of asylum over 1970–2023, we find that, averaging across years, some 87 percent of refugees originating from Sub-Saharan Africa were hosted in a country bordering their country of origin. Coverage is weaker in the early part of the period given that a sizable share of refugees is recorded without an identified country of origin so we treat this as suggestive rather than definitive.

because refugees exercise no choice, but because the feasible choice set is far more constrained and safety and accessibility receive substantially greater weight in the initial destination decision.

Nonetheless, we address this potential bias directly by adding measures of conflict intensity and natural disaster exposure to our specification. For conflict, we use the UCDP/PRIO Armed Conflict Dataset (Gleditsch et al., 2002; Uppsala Conflict Data Program and Peace Research Institute Oslo, 2026). The Armed Conflict Dataset records, at the country–year level, all armed conflicts involving a state as one of the primary parties that reach a threshold of at least 25 battle-related deaths in a calendar year, and classifies each conflict by type (interstate, intrastate, internationalized intrastate, or extrasystemic) and by intensity, distinguishing minor conflicts, with 25 to 999 battle-related deaths, from wars, with 1,000 or more. Our host-country measure is the resulting intensity level, taking the value of the most intense conflict active in the country-year. We prefer this measure to a continuous count of battle-related deaths for reasons of coverage.[13] Using battle-death counts would have required dropping the first two decades of the panel, a period that includes a substantial share of the displacement episodes we study. The intensity level trades some granularity for complete coverage, and preserves the distinction between minor conflicts and wars that is most relevant for our purposes. For natural disasters we use EM-DAT, the Emergency Events Database maintained by the Centre for Research on the Epidemiology of Disasters at UCLouvain (Delforge et al., 2025; CRED / UCLouvain, 2026). An event enters EM-DAT if it meets at least one of four criteria: ten or more reported deaths, one hundred or more people reported affected, the declaration of a state of emergency, or a call for international assistance. We restrict attention to natural hazards, excluding technological disasters, and construct a binary indicator for the occurrence of at least one qualifying event in a country–year. Both these controls enter the specification with a one-year lag, so that they are predetermined with respect to the refugee shock and cannot themselves be outcomes of it. This addresses the concern that host-country security conditions and disaster exposure jointly determine where displaced populations arrive and how the host economy performs: a country experiencing its own conflict is both a less feasible destination and, independently, likely to record weaker macroeconomic outcomes, so that failing to condition on host conditions would attribute to the refugee shock effects that originate elsewhere.

Before showing the results of the estimation we conduct a balance test shown in Table 6, which reports pre-treatment balance on these characteristics. Treated and control observations are statistically indistinguishable on conflict exposure at every margin: 22.9 percent of treated country-years record an armed conflict in the year before the shock against 23.4 percent of controls, and the difference net of country and year fixed effects is small and insignificant (0.035, $p = 0.40$). The same holds separately for minor conflicts and for wars,

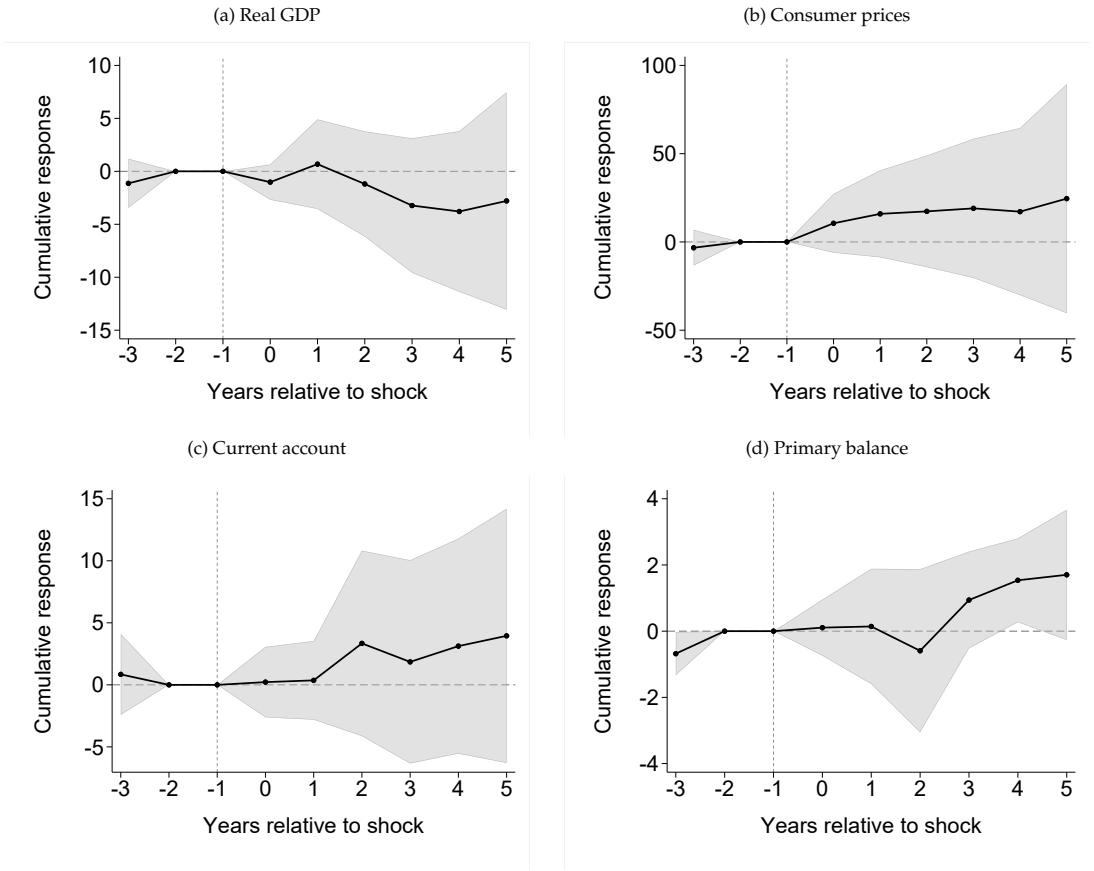
[13] The UCDP Georeferenced Event Dataset (Sundberg and Melander, 2013; Uppsala Conflict Data Program, 2026), from which such counts are constructed, begins in 1989, whereas the Armed Conflict Dataset extends back to 1946 and so spans our full 1970–2023 sample

and over the full five-year pre-window. We therefore do not find that our treated group and control group differ on their likelihood of conflict. Disaster exposure is balanced over the five-year window (82.4 percent against 92.9 percent, difference $-0.059$, $p = 0.29$), though treated country-years are less likely to record an event in $t - 1$ specifically.

Figure 9 and Figure 10 show the results of the estimation including the controls.[14] Adding the two controls leaves the results substantively unchanged. In the baseline specification (Table 11), the real GDP response remains statistically insignificant at every horizon, with point estimates somewhat more negative than in the baseline ($-2.79$ against $-0.87$ percentage points at $h = 5$) but confidence bands too wide to distinguish the two from 0. Consumer prices and the current account are essentially identical. The primary balance response remains similar though is now statistically significant at $h = 4$ while remaining indistinguishable from 0 at other horizons.

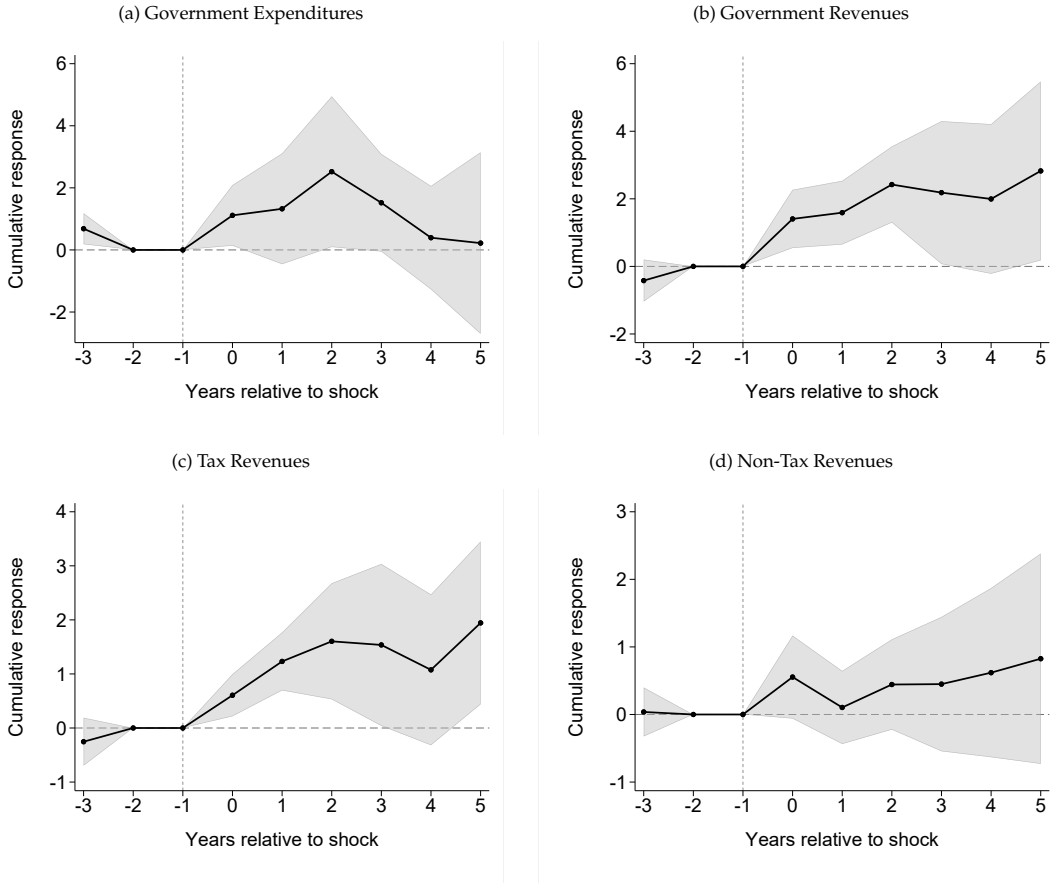
[14] The corresponding coefficients can be found in Tables 11 and 12.

Figure 9: LP-DiD Baseline with Conflict Intensity and Natural Disaster Covariates




*Note:* The figure reports the estimated dynamic effects of refugee inflow shocks using the baseline LP-DiD specification in Equation (1) on the log of real GDP × 100, log of headline CPI × 100, Current Account (percent of GDP) and Primary Balance (percent of GDP) with controls for host-country conflict intensity (UCDP/PRIO) and natural disaster exposure (EM-DAT), both entered with a one-year lag so as to be predetermined with respect to the shock. The solid lines show the point estimates and the shaded areas indicate 90 percent confidence intervals. Horizon $t-1$ is set to 0.

Figure 10: LP-DiD Estimates on the Fiscal Impacts with Conflict Intensity and Natural Disaster Covariates




*Note:* The figure reports the estimated dynamic effects of refugee inflow shocks using the baseline LP-DiD specification in Equation (1) on the government expenditure (excluding interest payments), government revenue, tax revenue and non-tax revenue all as a share of GDP. The solid lines show the point estimates and the shaded areas indicate 90 percent confidence intervals. Horizon $t-1$ is set to 0.

The fiscal mechanisms are more stable still (Table 12). Expenditures peak at 2.52 percentage points at $h = 2$ against 2.54 in the baseline; total revenues rise by 1.41 percentage points on impact against 1.37, and reaches 2.83 by $h = 5$ against 2.84; the tax revenue path is unchanged and remains significant at every horizon except $h = 4$. Non-tax revenue remains positive and insignificant throughout. Significance levels are unchanged everywhere except two marginal cases, expenditure at $h = 3$ and tax revenue on impact, each of which crosses a conventional threshold without any meaningful movement in the point estimate.

Taken together, these checks address the selection concern from two directions. The balance table shows that conflict, the factor most plausibly linking origin-country push factors to host-country outcomes, is uncorrelated with treatment in the variation our design uses, so there is no channel through which it could bias the estimates. The robustness estimates confirm this from the other side: conditioning on these characteristics leaves the coefficients essentially where they were. Since a control can only shift an estimate to the extent that it predicts both treatment and outcome, the combination of balance and coefficient stability is

stronger evidence than either alone. This is consistent with our reading of the destination-choice literature mainly that refugee flows in Sub-Saharan Africa are shaped principally by proximity, accessibility and safety, and the residual scope for sorting on host-country conditions is too limited to drive the responses we estimate.

A distinct concern is that the origin shocks driving our treatment may propagate geographically and contaminate the control group. If conflict in country A displaces people into a treated host, that same conflict may also reach a neighbor of A that remains in our control pool either through a smaller displacement inflow or through the direct cross-border transmission of violence. Our identifying assumptions cannot by themselves exclude contamination of this kind. We regard it as second-order, however, for three reasons. First, the scope for displacement-driven contamination is bounded by our own treatment definition: an inflow large enough to matter in macroeconomic terms would itself cross our threshold and enter the treated group, so the flows that leave a country in the control pool are by construction small and unlikely to carry a large macroeconomic footprint and our estimates are robust when using different treatment definitions in which we use different thresholds. Second, and most directly, we re-estimate our specification controlling for host-country conflict intensity as well as natural disasters, which absorbs the cross-border transmission of violence to potential controls; our results are essentially unchanged. Third, our comparison pool moreover extends beyond the immediate conflict neighborhood, so that many controls are geographically insulated from spillovers of a given origin shock.

*4.1.3. Extension: total-displacement as an alternative treatment definition*

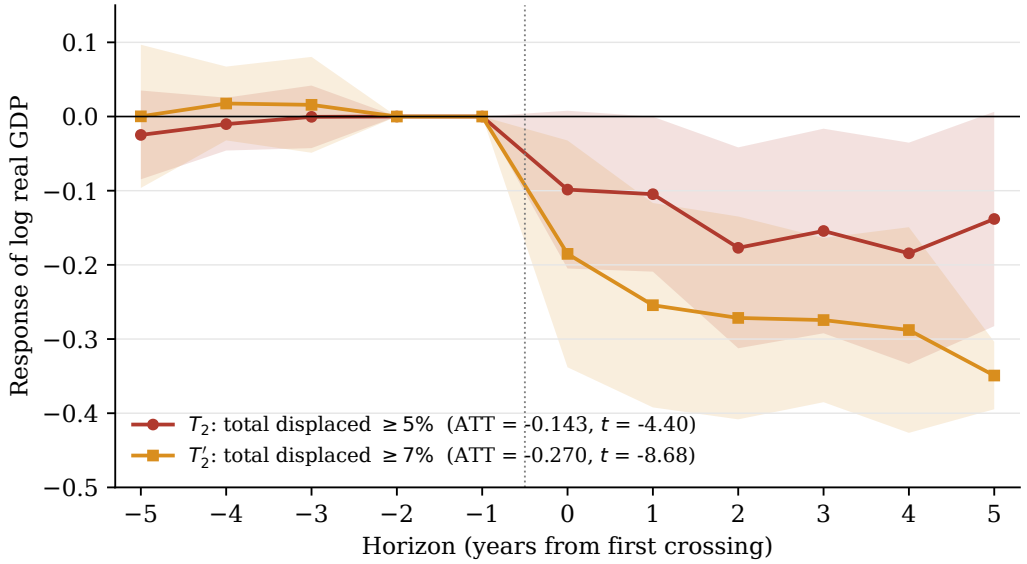
The previous sections focused on the macroeconomic effects of refugee inflow shocks. However, the stylized facts in Section 2 show that internally displaced persons (IDPs) now dominate displacement in SSA: by 2021, IDPs accounted for roughly three-quarters of all forcibly displaced persons in the region. This subsection therefore reports a complementary specification that broadens the treatment definition to include the total displaced population.

We define an alternative treatment, $T_2$, under which a country is treated from the first year in which its total displaced share (refugees plus IDPs) exceeds 5% of the host population; sensitivity to this choice is assessed using a 7% threshold ($T_2'$). The LP-DiD specification, fixed-effects structure, lag controls, and clustering are unchanged relative to the headline results; only the treatment indicator differs. The headline shock definition (refugees only, detected via a $Z$-score rule) and the total-displacement definition answer different policy-relevant questions: the former captures the effect of a discrete refugee inflow on the host economy, whereas the latter captures the effect of being displacement-affected.

Figure 11 reports LP-DiD impulse responses for log real GDP under both $T_2$ and $T_2'$, with 90% confidence intervals. The estimated effect is significantly negative under both definitions and accumulates monotonically

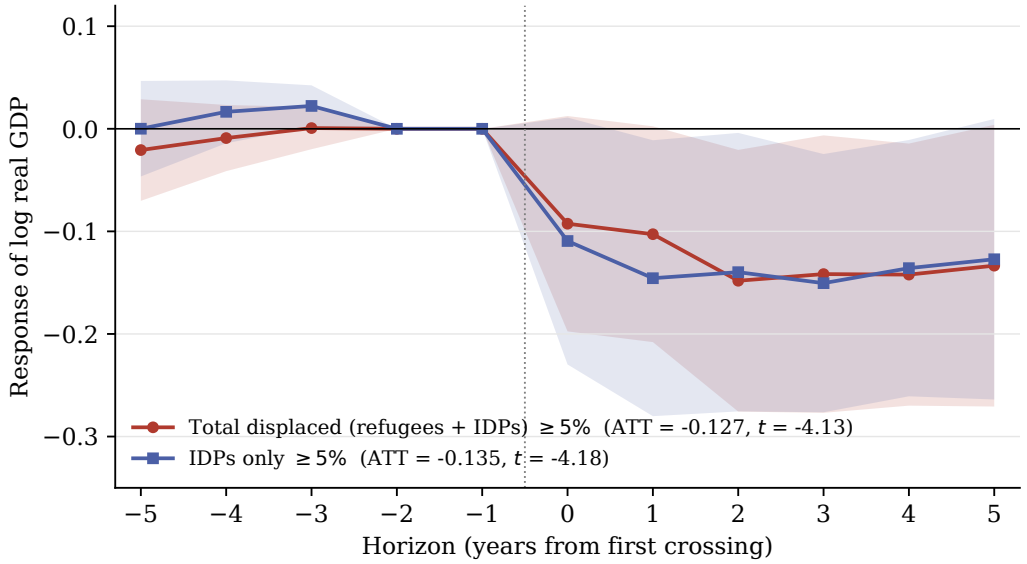
over the five-year horizon. Table 13 reports the corresponding pooled five-year ATTs: −0.143 ($t = -4.40$) for $T_2$ and −0.270 ($t = -8.68$) for $T_2'$. The dose–response between 5% and 7% is consistent with non-linear macroeconomic costs that intensify with displacement severity.

Figure 11: LP-DiD impulse response on log real GDP under total-displacement treatment definitions




A key identification caveat is in order. Countries that cross the $T_2$ threshold are disproportionately those experiencing active or recent conflict (e.g., the Democratic Republic of Congo, Central African Republic, Burkina Faso, South Sudan, Mali, and Mozambique). The estimated $T_2$ coefficient therefore conflates two effects that this cross-country design cannot cleanly separate: the macroeconomic effect of *being displacement-affected* and the macroeconomic effect of *the conflict that generates displacement*. The lagged-conflict control in the LP-DiD specification partially absorbs the conflict channel, but identification remains limited given the tight correlation between large IDP shares and the contemporaneous conflict environment. With that caveat in mind, Figure 12 compares the growth effects of the IDP treatment with those of total displacement. The results indicate that both measures are associated with broadly similar negative effects on GDP growth. As such, the $T_2$ estimate should therefore be interpreted as an upper bound on the pure displacement effect. This identification challenge motivates the country case studies in the following subsections, where higher-frequency within-country variation provides a relatively cleaner source of identification.

Figure 12: LP-DiD impulse response on log real GDP under total-displacement versus IDPs only




Overall, the $T_2$ extension corroborates the *direction* of the GDP effect under a broader treatment definition, while it cannot cleanly identify its *magnitude*. Reporting both specifications—the discrete refugee inflow shock and the total-displacement treatment—broadens the paper's coverage of the heterogeneous displacement landscape in Sub-Saharan Africa.

### *4.2. Micro Evidence of Forced Displacement Impacts*

#### *4.2.1. Burkina Faso*

This case study estimates the effects of internal displacement on host municipalities in Burkina Faso using the two-wave DiD framework with propensity score matching described in Section 3.1.3, applied to the household-survey panel introduced in Section 3.2.3. We examine four outcomes: wages, labor force participation, household consumption per capita, and food prices.

Results in Table 15 indicate no statistically significant aggregate wage effect of IDP inflows, although there is sectoral heterogeneity. IDP inflows are associated with lower wages for services workers, but upward wage pressure in industry, plausibly reflecting increased demand for skilled labor in host communities[15]. Overall, these findings suggest a largely stable wage structure alongside distributional effects across sectors. Consistent with expectations, education and work experience remain strong positive predictors of wages across specifications, underscoring the role of human capital in mitigating adverse labor market shocks

[15] For instance, demand for skilled labor in host communities could rise to meet infrastructure needs associated with accommodating IDPs.

(Mincer, 1974).

IDP inflows are also associated with lower employment opportunities (Table 16). The estimates imply a negative effect on employment probability, consistent with displacement shocks tightening labor market conditions in host communities. The literature similarly emphasizes that forced displacement can increase local labor supply while job creation remains limited, reducing employment prospects for the host population (Verme and Schuettler, 2021; George and Adelaja, 2022). This adverse impact is particularly pronounced in urban areas, where displaced households tend to cluster and competition for jobs is stronger.

Consistent with Table 17, IDP inflows generally do not significantly affect consumption per capita, although consumption falls for individuals working in agriculture and services. Higher demand for food and non-food products may exert upward pressure on prices, thereby decreasing consumption in these two sectors. In addition, since displaced households often engage in agricultural activities in host communities, the increase in labor supply may reduce productivity and translate into lower per capita consumption. Moreover, the 2018 and 2021 household surveys indicate that roughly 25 percent of Burkina Faso's population works in services, and about 42 percent of services workers hold wage employment (versus 22 percent across all sectors). This higher concentration of wage employment suggests greater exposure to wage shocks, which may contribute to the observed consumption declines among services households.

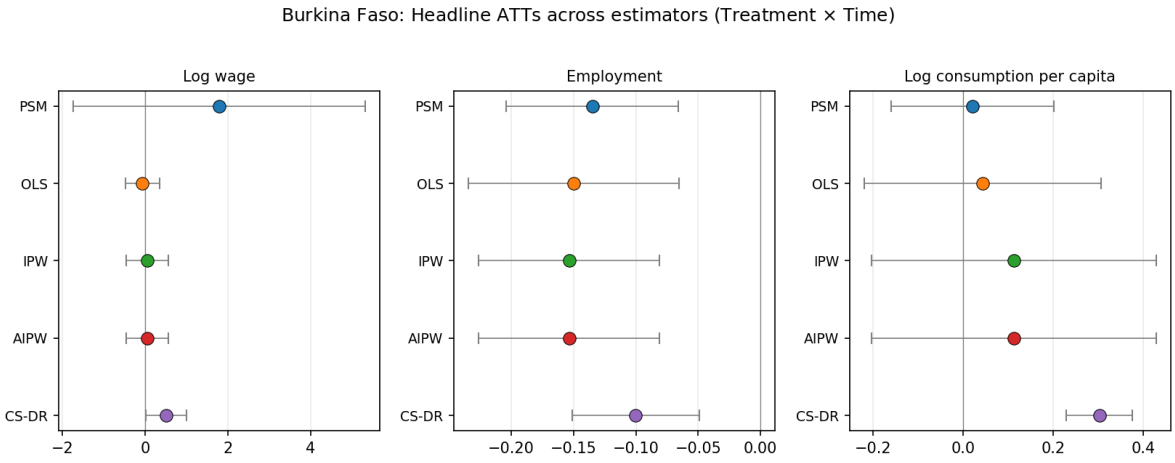
Moreover, we find no clear evidence that IDP inflows are associated with food inflation in host communities (Table 18). One plausible mechanism is that many IDPs engage in agriculture, contributing to local food production and stabilizing supply where they settle. Related evidence indicates that agriculture is a primary livelihood strategy among IDPs and can strengthen local supply and reduce vulnerability to food insecurity (FAO, 2024). IDPs' participation in farming may therefore mitigate food insecurity and limit inflationary pressures by increasing local production. A product-level decomposition for nine commodities, analogous to the South Sudan exercise reported in Section 4.2.3, yields directional patterns consistent with the imported-versus-local split (slightly lower local-cereal prices and slightly higher imported-rice prices), but limited regional coverage in the Burkina Faso dataset makes the design underpowered for commodity-specific inference and neither effect is statistically significant.

Finally, we conduct two robustness checks: alternative weighting schemes (*estimator robustness.*) and a proximity-based assessment of spillovers to control municipalities (*sample-restriction robustness.*).

*Estimator robustness.* The headline findings are broadly robust to alternative weights. Figure 13 reports the Treatment × Time coefficient for each headline outcome under five estimators: the baseline PSM specification, OLS, inverse-probability weighting (IPW), augmented IPW (AIPW), and the doubly robust

Callaway–Sant'Anna estimator. The results remain comparable to the baseline: aggregate wages and consumption are statistically indistinguishable from zero, while employment declines. Sectoral wage estimates remain statistically insignificant in most sectors and broadly comparable to the PSM-based results, although the industry and services coefficients are more sensitive to the weighting choice.

Figure 13: BF: headline ATTs across five estimators

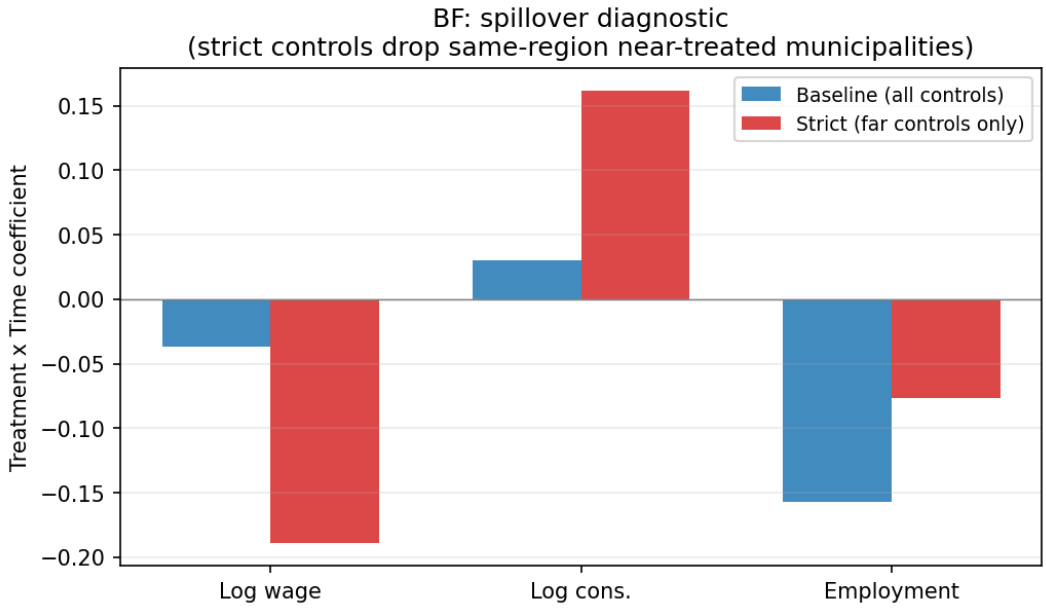



*Sample-restriction robustness.* A second robustness concern is potential contamination of the control group: some control municipalities border treated municipalities and may experience indirect, IDP-driven shocks to labor and goods markets, biasing the DiD estimates. We address this by restricting the control group to municipalities in regions without any treated municipality ("strict" control). Figure 14 compares baseline and strict-control ATTs for employment, wages, and consumption. The largest revision concerns employment: the coefficient declines in magnitude from −0.157 (baseline, all controls) to −0.077 (strict, far controls only), suggesting that the baseline employment effect partly reflects spillovers to nearby control areas.

*Identification credibility.* Parallel-trends testing is not directly available with the two-wave panel design; identification credibility instead rests on the propensity-score-matching and threshold-restriction choices documented in Section 3.1.4, which collectively mitigate selection on observables between treated and control municipalities.

Overall, this case study provides nuanced evidence on the economic effects of internal displacement in Burkina Faso. IDP inflows are associated with lower services-sector earnings and reduced labor force participation, particularly in urban areas. The consumption declines observed among agricultural and services workers further underscore household vulnerability to displacement shocks. At the same time, the absence of significant food price inflation is consistent with IDPs' engagement in agriculture stabilizing local

Figure 14: BF: spillover diagnostic on baseline versus strict controls




supply. These findings point to the need for targeted policies that support job creation in host communities while leveraging IDPs' contributions to local labor markets and agricultural production to strengthen food security.

*4.2.2. Uganda*

This case study estimates the effects of refugee inflows on host districts in Uganda using the two-wave DiD framework described in Section 3.1.3 (without propensity score matching), applied to the multi-wave LSMS panel introduced in Section 3.2.4. We examine four outcomes: wages, labor force participation, household consumption per capita, and inflation, with emphasis on food prices.

The econometric results in Table 21 indicate no statistically significant effects of refugee inflows on wages in host communities. This may reflect the fact that refugees are primarily located in designated settlements rather than fully integrated into local labor markets. Although refugees are legally granted freedom of movement in Uganda, in practice their economic activity appears to remain largely concentrated within settlements.

Refugee inflows, however, are associated with a higher probability of employment among host-community members, particularly in rural areas where refugee settlements are concentrated (Table 22). This pattern suggests that refugee arrivals may generate local employment opportunities, especially where public authorities and humanitarian organizations expand operations to deliver assistance. Rather than indicating broad-based improvements in labor market conditions, the results are consistent with localized labor-demand

effects associated with the humanitarian presence. In addition, because refugee inflows do not have any significant effects on wages, employment gains are likely concentrated in self-employment and other non-wage activities.

Evidence from Table 23 further indicates that refugee inflows are associated with lower household consumption per capita in host communities, regardless of sector of activity or place of residence. Despite potential demand stimulus from humanitarian assistance and increased local spending, the net welfare effect for host households appears negative. One plausible mechanism is that refugee arrivals increase pressure on local goods and essential services, including housing, transport, health care, and other basic facilities—thereby tightening budget constraints and reducing consumption as prices rise or access becomes more limited. [16]

Finally, refugee inflows are associated with inflationary effects that vary across product categories (Table 24). While prices rise for some processed food items (e.g., sugar and tea), prices fall for certain locally produced staples (e.g., corn and rice), potentially reflecting refugees' contribution to agricultural production. These patterns suggest that market adjustment to refugee inflows depends on product-specific supply conditions and the speed with which traders respond to demand shocks. Overall, the results underscore the importance of examining disaggregated price dynamics rather than relying exclusively on aggregate inflation measures.

*Sample-restriction robustness.* As a robustness check, we re-estimate the headline DiD on the most recent two waves of the LSMS-ISA panel (2015/16 and 2018/19), under two alternative treatment specifications: a binary indicator for settlement-hosting districts (constructed directly from UNHCR `demographics_residing` data), and a continuous refugee-share regressor interacted with the post-2016 dummy. Figure 15 reports the resulting coefficients with 90% confidence intervals for Treatment × Time for the three core outcomes. The binary settlement-district specification produces statistically null effects on all three outcomes (wages, employment,consumption), reflecting the coarseness of the binary indicator relative to the 0.5% refugee-share treatment. The continuous-dose specification, however, reproduces the qualitative direction of the results in Table 22 employment finding. It also shows a statistically significant downward pressure on wages, consistent with labor-supply expansion in settlement-hosting districts. The consumption interaction is marginally positive on the post-2015 window, somewhat softer than those in Table 23 negative.[17]

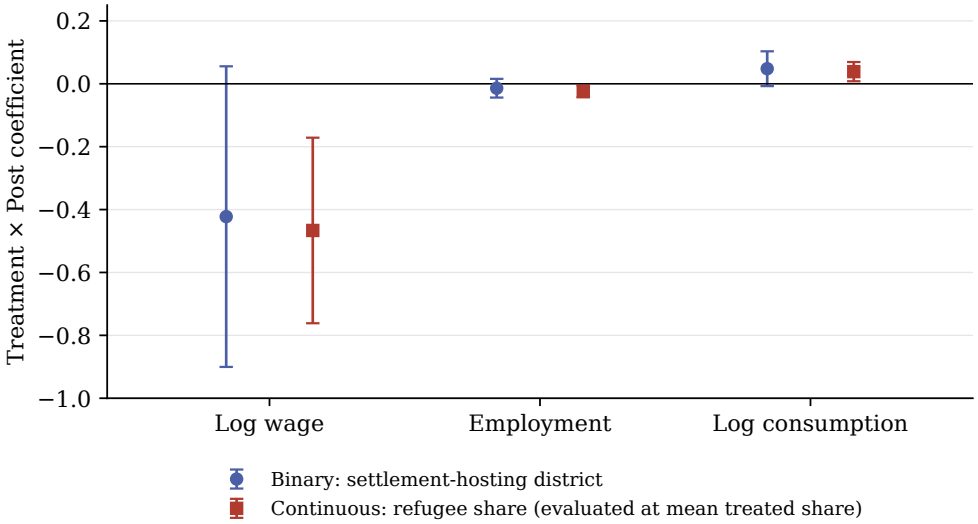
*Identification credibility.* As in the Burkina Faso case study, parallel-trends testing is not directly available with the two-wave panel design; identification credibility instead rests on the threshold-restriction choice documented in Section 3.1.4. Propensity-score matching is not applied because, in the Uganda setting,

[16] This finding is consistent with Verme and Schuettler (2021) systematic review, which shows that although the welfare effects of forced displacement are mixed, approximately one-fifth of studies report adverse welfare effects on host communities, often linked to increased pressure on local goods, services, and other scarce resources

[17] The post-2015 panel is the subset of the LSMS-ISA Uganda surveys for which household-level identifiers harmonize cleanly.

treatment assignment is only weakly correlated with observable host-community characteristics.

Figure 15: Uganda: binary versus continuous treatment specification on the post-2015 LSMS window




Overall, we find that refugee inflows have nuanced effects on host communities in Uganda. While they do not significantly affect wages, refugee inflows are associated with increased employment opportunities, likely driven by the expansion of humanitarian activities. At the same time, pressures on local goods and services have a net negative effect on household consumption, while the effect on food prices is heterogeneous across food categories.

#### *4.2.3. South Sudan*

This case study estimates the effects of the April 2023 displacement shock on consumer food prices in South Sudan using the sub-national LP-DiD framework described in Section 3.1.2, applied to the county-month RTFP panel introduced in Section 3.2.2. We examine three displacement-inflow series in turn: total inflows (refugees, asylum-seekers, and returnees combined), refugees and asylum-seekers only, and returnees only. Unlike the Burkina Faso and Uganda case studies, which exploit multi-outcome household-survey data on wages, labor force participation, consumption, and inflation, the South Sudan analysis focuses exclusively on consumer food prices: the World Bank RTFP database(Andrée, 2026) provides monthly market-level price variation at the frequency required for a local-projections design, which the annual LSMS surveys cannot offer[18].

[18] An important caveat is that RTFP prices are not all directly observed. The database combines market price surveys from WFP, FAO, and national statistical sources with machine-learning imputation of missing observations (Andrée, 2021). In South Sudan, where market surveys are frequently interrupted by insecurity and access constraints, a non-trivial share of county-month observations may therefore be model-based estimates rather than recorded prices. To the extent that imputation smooths local price movements or borrows information from neighboring markets and periods, it is likely to attenuate estimated responses toward zero and to weaken

*Results for shorter horizons*

At the three-month post-shock horizon, the aggregate-inflows ATT is small and statistically indistinguishable from zero (Fig. 16; pooled coefficient 0.012, $t = 0.72$): prices tend to increase on impact but the effects fizzle out within a month. When we disaggregate by group, refugees-only inflows show a persistent but imprecise positive trend of 1.7% on impact and remains statistically indistinguishable from zero throughout the three-month window after the shock; returnees-only inflows show a positive impact-month effect ($\hat{\beta} = 0.059$, $t = 2.63$; +6.1%) that dissipates within two months. A natural interpretation is that returnees re-integrate into host communities faster than refugees and asylum-seekers, enabling them to resume subsistence agriculture and ameliorating upward price pressure. These short-run findings are robust to bounded pre-trend deviations under the framework of Rambachan and Roth (2023): under the most stringent $M = 0$ calibration, the robust 95% confidence interval for the returnees group at impact is $[0.06, 0.25]$, while for the aggregate group it spans zero.[19]

*Effects at longer horizons*

Restricting attention to the three-month horizon, however, obscures a markedly different medium-run picture. Extending the post-shock window to twelve months (`pre(3) post(12)`) reveals a horizon path that deepens over time, reaching about −14.8% by month 12 ($\hat{\beta}_{h12} = -0.160$, $t = -3.14$; Fig. 16). The returnees-only and refugees-only series move in the same direction. This delayed deflationary effect is consistent with two reinforcing channels: returnees resume agricultural production over the first post-shock year, expanding local food supply; and in-kind humanitarian food deliveries and cash-transfer operations ramp up progressively, depressing local prices in treated counties.

*Product-level heterogeneity (Mechanism):*

The averaged food-price index masks substantial within-category heterogeneity. Re-estimating the LP-DiD specification separately for each of fifteen commodities in the WFP per-commodity South Sudan dataset reveals a sharp product-category split at the twelve-month horizon (Fig. 17). Imported and processed staples rise substantially: maize meal +68% ($t = +4.56$), wheat flour +32% ($t = +3.14$), and imported brown sugar +11% ($t = +2.28$). Local cereals stay flat or fall, with millet white at −19% ($t = -2.80$) the only significant negative. This pattern is consistent with humanitarian aid substituting more directly for the local-cereal basket than for imported and processed staples, on which displaced and host households place additional demand—an interpretation that aligns with the established literature on refugee inflows and host-community prices in SSA (Verme and Schuettler, 2021; Maystadt and Verwimp, 2014; Alix-Garcia and

the contrast between treated and control counties. Our estimates should accordingly be read as conservative, and interpreted with this measurement limitation in mind.

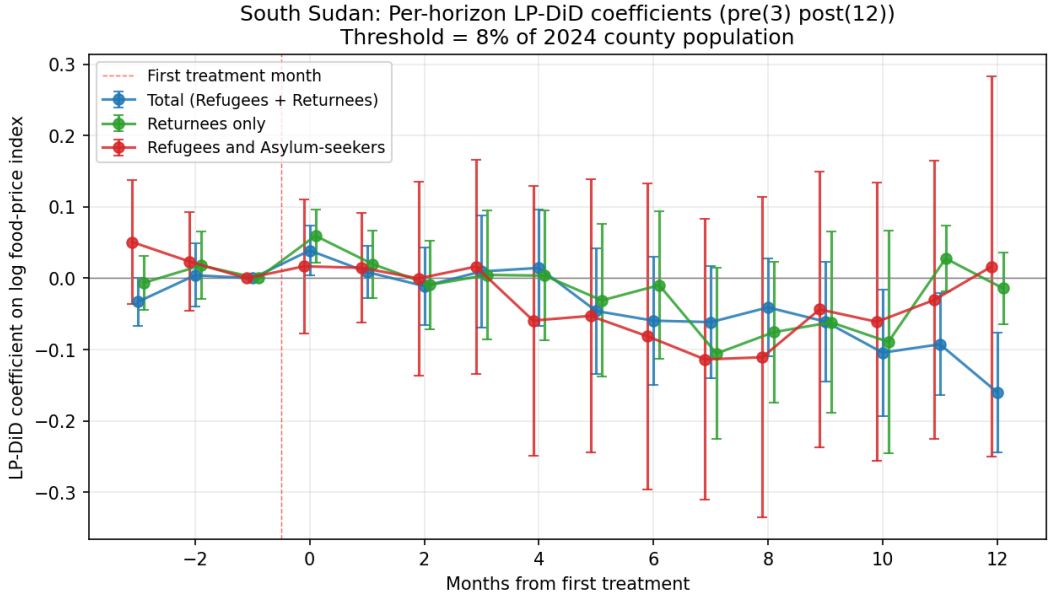
[19] The available pre-shock window is only three months, so the credible DiD bounds are mechanically loose; a longer pre-period would tighten them.

Saah, 2010).

*Sample-restriction robustness (spatial spillover).*
A natural concern is that the impact-horizon null reflects contamination of the control group by spatial spillovers across neighboring counties. Restricting the control set to counties not adjacent (via shared state membership) to any treated county yields a strict-control ATT of +0.007 versus the baseline +0.012: a 0.5% shift well within sampling noise. A complementary near-versus-far placebo on the refugees-only subsample finds that prices in near-controls actually *fell* relative to far-controls ($t = -2.05$)—opposite to a demand-driven contamination pattern, and consistent with humanitarian aid spilling onto neighboring counties, reinforcing the aid-offset interpretation.

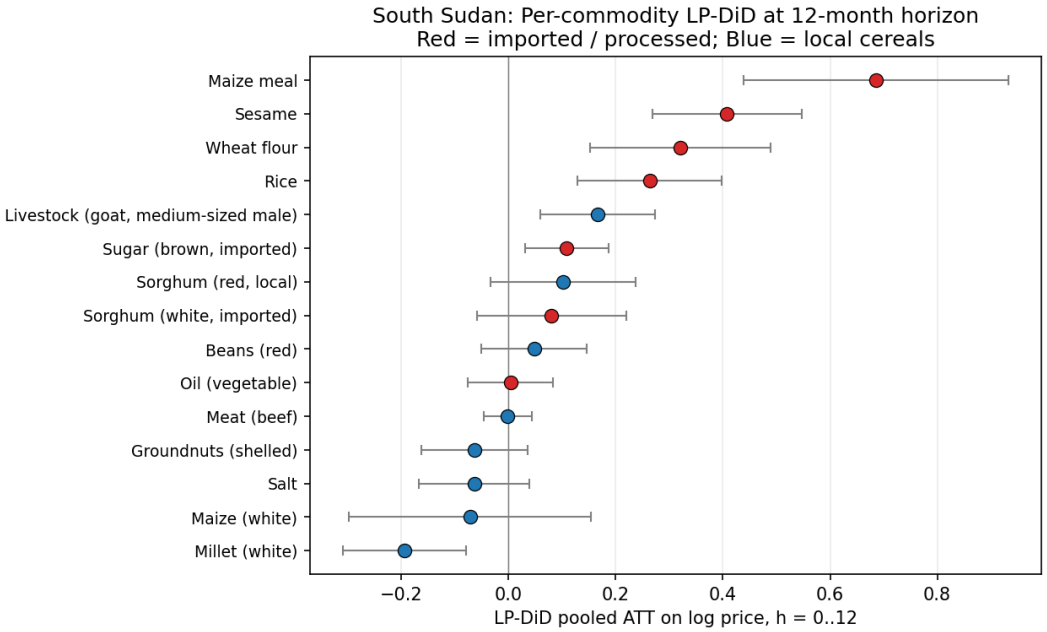
Overall, the South Sudan evidence indicates that, at shorter horizons, forced displacement leaves the averaged food-price index essentially unchanged. At the twelve-month horizon, however, treated counties show a small but precisely-estimated decline in average prices—a decline that, at the product level, masks large and divergent movements: imported and processed staples rise substantially while most local cereals stay flat or fall. The integrating mechanism is the dual role of returnees as resumed producers of local staples and of humanitarian aid as a price-stabilizer for the same basket, leaving the imported-and-processed segment to absorb the residual demand pressure.

Figure 16: South Sudan: Food Price Impact at Twelve-Month Horizon




*Note*: Per-horizon LP-DiD coefficients (90% CI shaded), `pre(3) post(12)`, threshold = 8% of 2024 county population. The aggregate-inflows series transitions from a near-zero impact at $h = 3$ to a statistically significant negative effect by $h = 12$.

Figure 17: South Sudan: Per-Commodity LP-DiD ATT at Twelve-Month Horizon




*Note*: Plot of per-commodity LP-DiD pooled ATTs with 90% CI, `pre(3) post(12)`. Imported and processed staples (maize meal, sesame, wheat flour, rice, sugar) rise substantially; local cereals stay flat or fall (millet white is the only significant negative).

### *4.3. Cross-case findings*

*Food-price mechanism.* A pattern worth noting across the three case studies is the consistency of the food-price decomposition: prices of imported and processed staples rise more than (or relative to) prices of locally produced cereals. In Table 24, sugar ($+0.142^{***}$) and tea ($+0.981^{***}$) rise while corn ($-0.070^{**}$) and rice ($-0.047^{*}$) fall. The South Sudan per-commodity decomposition (Section 4.2.3) and the Burkina Faso per-commodity analysis (also reported above) both display the same directional pattern, suggesting that the imported-versus-local split is a robust feature of host-community food-price response to forced displacement in Sub-Saharan Africa, even as the underlying conflict context differs across the three cases.

*Robustness across cases.* Taken together, the robustness battery summarized in Table 3 supports the main findings of this paper. On *identification*, pre-trend coefficients in the cross-country LP-DiD are statistically indistinguishable from zero, and the short-run South Sudan estimates survive bounded pre-trend deviations under the credible DiD framework of Rambachan and Roth (2023). The two-wave designs in Burkina Faso and Uganda rest instead on the matching and threshold-restriction choices documented in Section 3.1.4. On *estimator choice*, the Burkina Faso findings are broadly stable across five estimator families (PSM, OLS, IPW, AIPW, and Callaway-Sant'Anna), with the largest movement on the employment coefficient. The other analyses do not currently include an estimator-robustness layer. On *sample restriction and treatment definition*, the cross-country result is unchanged when restricting controls to SSA only and survives a broader total-displacement treatment definition. The Burkina Faso employment decline halves when restricting to

non-adjacent "far" controls, suggesting that part of the baseline effect reflects spatial spillover. The Uganda employment finding is sensitive to whether the 0.5% refugee-share is implemented as a binary or continuous treatment. The South Sudan impact-horizon null is unchanged when treated counties' neighbors are excluded from the control set. Overall, the directional findings of the paper—limited macroeconomic effects on national-level GDP and inflation, localized but heterogeneous labor-market effects, and a consistent imported-versus-local food-price decomposition—are stable across the battery of robustness checks.

## 5. Conclusion

This paper provides the first systematic cross-country analysis of the macroeconomic effects of forced displacement in Sub-Saharan Africa, employing a difference-in-differences methodology. Panel regressions using macroeconomic data are complemented by case studies of Burkina Faso, South Sudan, and Uganda that add granularity to the analysis. The results point to a consistent pattern: large inflows of forcibly displaced persons are associated with muted effects on national-level output, prices, and external balances. The effects on the primary fiscal balance are not statistically significant in general, but this masks partially offsetting movements: an increase in government expenditures is offset by persistently higher revenue. Localized impacts on labor markets and consumption, however, are more heterogeneous.

The case studies largely corroborate these aggregate findings on inflation while providing additional nuance. In South Sudan, displacement leaves the average food-price index essentially unchanged at shorter horizons but produces a small, precisely estimated decline at the twelve-month horizon; this average, in turn, masks large and divergent product-level movements, with imported and processed staples rising substantially while locally produced cereals stay flat or fall. Similarly, difference-in-differences analysis using household survey data from Burkina Faso and Uganda indicates that large inflows of forcibly displaced persons do not lead to economically meaningful changes in aggregate food inflation in host communities. This imported-versus-local split is a robust feature across all three cases and points to an important mitigating mechanism: IDPs and refugees often engage in agricultural activities, thereby contributing to local food production and supporting food supply in areas where they settle, even as additional demand pushes up the prices of imported and processed goods.

Moreover, at the local level, the evidence points to more complex and context-dependent outcomes. In Burkina Faso, inflows of internally displaced persons are associated with upward wage pressures in the industrial sector and downward pressures in the services sector, alongside reduced labor force participation in host communities—particularly in urban areas. In Uganda, refugee inflows do not have a statistically significant effect on wages, but are associated with increased employment opportunities in host communities. Demand pressures on local goods and services may constrain household consumption, while the effects on food prices remain heterogeneous across product categories.

Evidence that IDPs and refugees contribute to agricultural production underscores the importance of facilitating their access to land, agricultural inputs, and extension services. Such policies not only support the livelihoods of displaced populations but can also generate positive spillovers for host communities by stabilizing food supply and prices. Finally, given the persistent and large-scale nature of forced displacement in SSA, policies that unlock the productive potential of displaced populations and facilitate their socioeconomic integration are warranted.

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Table 1: Expected Effects of Forced Displacement on Host Countries

| Short Run | Medium/Long Run |
|---|---|
| **Growth and Inflation** | |
| Higher labor supply and domestic demand; price pressures; heightened uncertainty? | Higher growth and productivity if FDPs are integrated; misallocation absent enabling conditions. |
| **Fiscal** | |
| Higher public spending; wider deficits or re-prioritization; crowding-out of public investment? | Larger tax base (with enabling conditions) ⇒ easing of fiscal pressures. |
| **External Sector** | |
| Weaker current account balance; real exchange rate appreciation and lower competitiveness | |
| **Labor Market** | |
| Higher labor supply and informality ⇒ downward wage pressures. | |

Table 2: Design choices across the four analyses

| Analysis | Estimator | PSM | Treatment variable | Threshold |
|---|---|---|---|---|
| Cross-country (Section 3.1.1) | LP-DiD | No | Net refugee inflow / population (annual) | $z > 1.5$ & $> 0.5\%$ pop. |
| South Sudan (Section 3.1.2) | LP-DiD | No | Cumulative refugees + returnees / county pop. | $\geq 8\%$ |
| Burkina Faso (Section 3.1.3) | Two-wave DiD | Yes | IDP share of host municipality | $\geq 0.05\%$ |
| Uganda (Section 3.1.3) | Two-wave DiD | No | Refugee share of host district | $\geq 0.5\%$ |

Table 3: Robustness checks across the four analyses

| Analysis | Identification | Estimator | Sample / treatment definition |
|---|---|---|---|
| Cross-country | Pre-trends $h \in [-5, -2]$ | — | SSA-only restriction; $T_2$ total-displacement |
| South Sudan | Honest DiD bounds (Rambachan-Roth) | — | Strict-control (spatial spillover) |
| Burkina Faso | — *(two-wave design)* | PSM, OLS, IPW, AIPW, CS | Strict-control (spatial spillover) |
| Uganda | — *(two-wave design)* | — | Binary vs. continuous treatment |

*Note*: Each cell lists the robustness test(s) reported for that analysis within each of three taxonomic categories. "—" indicates the test is not currently reported. The two-wave DiD design for Burkina Faso and Uganda does not support direct pre-trend testing.

Table 4: Country-Year Shocks (Treated Group)

| Country | ISO | Year | Z-score | Country | ISO | Year | Z-score |
|---|---|---|---|---|---|---|---|
| Angola | AGO | 1977 | 3.17 | Lesotho | LSO | 1980 | 4.09 |
| Benin | BEN | 1993 | 4.22 | Liberia | LBR | 1992 | 2.62 |
| Botswana | BWA | 1978 | 3.85 | Malawi | MWI | 1987, 1988, 1989 | 3.70, 2.27, 1.64 |
| Burundi | BDI | 1980 | 4.98 | Mauritania | MRT | 1990, 2012 | 2.86, 1.62 |
| Cameroon | CMR | 1980, 2014 | 3.88, 1.77 | Mozambique | MOZ | 1978 | 3.90 |
| C. Afr. Rep. | CAF | 1984, 2023 | 2.86, 1.51 | Namibia | NAM | 2000 | 4.46 |
| Chad | TCD | 2003, 2004 | 2.66, 2.40 | Rwanda | RWA | 1993 | 4.70 |
| Congo, Rep. | COG | 2000, 2009 | 3.41, 2.53 | Senegal | SEN | 1989 | 1.98 |
| Congo, D.R. | COD | 1994 | 5.16 | Sierra Leone | SLE | 1990 | 3.13 |
| Côte d'Ivoire | CIV | 1975, 1976 | 3.19, 1.73 | Somalia | SOM | 1979, 1980 | 2.74, 3.76 |
| Djibouti | DJI | 1989, 1990 | 2.07, 2.86 | South Africa | ZAF | 2015 | 3.43 |
| Eswatini | SWZ | 1979, 1988 | 1.74, 4.60 | South Sudan | SSD | 2012 | 2.53 |
| Ethiopia | ETH | 1988 | 3.87 | Sudan | SDN | 1984, 1985, 1990, 2017 | 2.82, 1.55, 2.20, 2.15 |
| Gabon | GAB | 1975 | 2.95 | Tanzania | TZA | 1972, 1993, 1994 | 1.76, 2.91, 3.21 |
| Ghana | GHA | 1993 | 3.70 | Uganda | UGA | 2016, 2017 | 3.80, 2.92 |
| Guinea | GIN | 1991 | 2.60 | Zambia | ZMB | 1977, 1982 | 2.11, 3.18 |
| Guinea-Bissau | GNB | 1992, 1993 | 1.92, 2.10 | Zimbabwe | ZWE | 1987 | 3.41 |
| Kenya | KEN | 1992 | 4.45 | | | | |

Table 5: Unconditional pre-trends, macroeconomic outcomes: LP-DiD placebo estimates (Full Sample)

| | Real GDP | Consumer prices | Current account | Primary balance |
|---|---|---|---|---|
| | (log pts × 100) | (log pts × 100) | (% of GDP) | (% of GDP) |
| Horizon | (1) | (2) | (3) | (4) |
| $h = -10$ | 11.692** | -2.012 | 1.072 | 0.459 |
| | ( 4.787) | (64.659) | ( 2.463) | ( 2.315) |
| $h = -9$ | 8.491* | -0.668 | 4.704 | 0.804 |
| | ( 4.391) | (52.273) | ( 4.331) | ( 2.019) |
| $h = -8$ | 3.635 | -15.447 | 1.213 | 3.868 |
| | ( 5.148) | (44.303) | ( 3.379) | ( 2.860) |
| $h = -7$ | 4.505 | -5.733 | 3.723 | 1.629 |
| | ( 4.497) | (39.321) | ( 3.644) | ( 1.095) |
| $h = -6$ | 3.516 | -9.747 | 2.695 | 0.707 |
| | ( 3.835) | (37.599) | ( 3.020) | ( 1.084) |
| $h = -5$ | 1.893 | -14.581 | 2.099 | -0.000 |
| | ( 3.074) | (33.583) | ( 2.907) | ( 1.100) |
| $h = -4$ | 0.045 | -20.676 | 2.603 | -0.530 |
| | ( 2.747) | (32.960) | ( 2.772) | ( 0.977) |
| $h = -3$ | -0.220 | -12.105 | 1.470 | -1.163** |
| | ( 2.064) | (20.935) | ( 2.594) | ( 0.571) |
| $h = -2$ | 0.372 | -4.581 | 0.767 | 0.100 |
| | ( 1.055) | ( 7.852) | ( 2.000) | ( 0.669) |
| $h = -1$ | 0 | 0 | 0 | 0 |
| | | [Normalization horizon] | | |
| *Pooled pre-period average* | | | | |
| $h \in [-5, -2]$ | 0.445 | -13.319 | 1.862 | -0.538 |
| | ( 2.187) | (23.986) | ( 2.462) | ( 0.709) |
| $h \in [-10, -2]$ | 9.332** | -16.103 | -1.386 | 0.196 |
| | ( 3.787) | (52.010) | ( 1.382) | ( 1.162) |
| Median $N$ | 2751 | 2727 | 2732 | 1934 |

*Notes:* Unconditional pre-treatment placebo estimates from the LP-DiD. Each coefficient comes from a regression of $y_{t-h} - y_{t-1}$ on the treatment indicator with country and year fixed effects only (no lagged outcomes or other controls), estimated on the corresponding clean-control sample. Columns (1) and (2) are in log points × 100; columns (3) and (4) in percentage points of GDP. Pooled rows report the coefficient from a single regression of the average of $y_{t-h} - y_{t-1}$ over the indicated horizons on the treatment indicator, with the same fixed effects, estimated on the clean-control sample of the earliest horizon in the window. Standard errors clustered at the country level in parentheses. $h = -1$ is the normalisation horizon and is zero by construction. $^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$. Figures report 90% confidence intervals.

Table 6: Pre-treatment balance on conflict and disaster exposure (Full Sample)

| | Treated mean (1) | Control mean (2) | Difference (net of FE) (3) | $p$-value (4) | $N$ (5) |
|---|---|---|---|---|---|
| *Panel A: State in the year before the shock ($t-1$)* | | | | | |
| Any armed conflict | 0.229 | 0.234 | 0.035 | 0.402 | 3600 |
| | | | ( 0.041) | | |
| Minor conflict (25–999 deaths) | 0.143 | 0.179 | 0.022 | 0.614 | 3600 |
| | | | ( 0.043) | | |
| War (1,000+ deaths) | 0.086 | 0.055 | 0.013 | 0.713 | 3600 |
| | | | ( 0.035) | | |
| Any natural disaster | 0.343 | 0.697 | -0.262*** | 0.000 | 3600 |
| | | | ( 0.069) | | |
| *Panel B: Any occurrence over $t-1$ to $t-5$* | | | | | |
| Any armed conflict | 0.265 | 0.360 | -0.068 | 0.204 | 3468 |
| | | | ( 0.053) | | |
| Minor conflict (25–999 deaths) | 0.206 | 0.333 | -0.078 | 0.195 | 3468 |
| | | | ( 0.060) | | |
| War (1,000+ deaths) | 0.176 | 0.111 | 0.032 | 0.478 | 3468 |
| | | | ( 0.046) | | |
| Any natural disaster | 0.824 | 0.929 | -0.059 | 0.288 | 3468 |
| | | | ( 0.055) | | |

*Notes:* The sample is the clean estimation sample at horizon $h = 0$: treated observations are country-years with a refugee inflow shock, and controls are clean control observations. All characteristics are measured before the shock year. Panel A reports the value in $t-1$; Panel B reports an indicator for any occurrence in the preceding 5 years (missing if the full window is not observed). Columns (1)–(2) report group means. Column (3) reports the coefficient from regressing each characteristic on treatment with country and year fixed effects, the variation used for identification; standard errors clustered by country in parentheses. Conflict data are from UCDP/PRIO (minor conflict: 25–999 battle-related deaths per year; war: 1,000 or more). Natural disasters are from EM-DAT (geophysical, hydrological, meteorological, biological, and climatological hazards).

$^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$.

Table 7: LP-DiD estimates: response to a refugee inflow shock (Full Sample)

| Horizon | Real GDP (log pts × 100) (1) | Consumer prices (log pts × 100) (2) | Current account (% of GDP) (3) | Primary balance (% of GDP) (4) |
|---|---|---|---|---|
| $h = -3$ | -0.853 | -3.907 | 0.847 | -0.924*** |
| | ( 1.236) | ( 6.052) | ( 1.832) | ( 0.302) |
| $h = -2$ | - | - | - | - |
| $h = -1$ | 0 | 0 | 0 | 0 |
| | | [Normalization horizon] | | |
| $h = 0$ | -0.661 | 9.452 | 0.273 | 0.195 |
| | ( 1.000) | ( 9.169) | ( 1.628) | ( 0.473) |
| $h = 1$ | 1.178 | 13.475 | 0.416 | 0.111 |
| | ( 2.357) | (12.513) | ( 1.833) | ( 0.866) |
| $h = 2$ | -0.122 | 14.006 | 3.034 | -0.415 |
| | ( 2.809) | (16.891) | ( 4.177) | ( 1.270) |
| $h = 3$ | -1.927 | 13.383 | 1.900 | 0.227 |
| | ( 3.512) | (21.432) | ( 4.564) | ( 0.833) |
| $h = 4$ | -2.254 | 10.031 | 3.187 | 0.944 |
| | ( 4.206) | (25.555) | ( 4.708) | ( 0.773) |
| $h = 5$ | -0.649 | 12.104 | 3.780 | 1.648 |
| | ( 5.527) | (34.022) | ( 5.396) | ( 1.070) |
| Median $N$ | 3062 | 3032 | 3040 | 2166 |

*Notes:* LP-DiD estimates include country and year fixed effects; standard errors clustered at the country level in parentheses. $h = -1$ is the normalisation horizon and is zero by construction. Columns (1) and (2) report cumulative responses in log points × 100; columns (3) and (4) report the response of the ratio to GDP in percentage points at horizon $h$, which is a level gap rather than a cumulative sum. $^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$. Figures report 90% confidence intervals.

Table 8: Fiscal mechanisms: response to a refugee inflow shock (Full Sample)

| | Expenditure (% of GDP) | Revenue (% of GDP) | Tax revenue (% of GDP) | Non-tax revenue (% of GDP) |
|---|---|---|---|---|
| Horizon | (1) | (2) | (3) | (4) |
| $h = -3$ | 0.727** | -0.346 | -0.215 | 0.010 |
| | ( 0.301) | ( 0.338) | ( 0.257) | ( 0.207) |
| $h = -2$ | - | - | -* | - |
| $h = -1$ | 0 | 0 | 0 | 0 |
| | | [Normalization horizon] | | |
| $h = 0$ | 1.076* | 1.367*** | 0.620*** | 0.508 |
| | ( 0.572) | ( 0.515) | ( 0.233) | ( 0.371) |
| $h = 1$ | 1.290 | 1.569*** | 1.242*** | 0.093 |
| | ( 1.052) | ( 0.562) | ( 0.320) | ( 0.322) |
| $h = 2$ | 2.536* | 2.406*** | 1.609** | 0.441 |
| | ( 1.426) | ( 0.679) | ( 0.636) | ( 0.398) |
| $h = 3$ | 1.526* | 2.178* | 1.532* | 0.450 |
| | ( 0.915) | ( 1.270) | ( 0.895) | ( 0.603) |
| $h = 4$ | 0.404 | 1.986 | 1.072 | 0.615 |
| | ( 0.988) | ( 1.332) | ( 0.827) | ( 0.753) |
| $h = 5$ | 0.269 | 2.837* | 1.914** | 0.844 |
| | ( 1.735) | ( 1.584) | ( 0.907) | ( 0.914) |
| Median $N$ | 2184 | 2282 | 1996 | 1992 |

*Notes:* LP-DiD estimates include country and year fixed effects; standard errors clustered at the country level in parentheses. $h = -1$ is the normalisation horizon and is zero by construction. All outcomes are expressed as a share of GDP; coefficients report the response in percentage points at horizon $h$. $^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$. Figures report 90% confidence intervals.

Table 9: LP-DiD estimates: response to a refugee inflow shock (SSA Countries Only)

| Horizon | Real GDP (log pts × 100) (1) | Consumer prices (log pts × 100) (2) | Current account (% of GDP) (3) | Primary balance (% of GDP) (4) |
|---|---|---|---|---|
| $h = -3$ | -1.357 | -6.740 | 0.542 | -1.536*** |
| | ( 1.081) | ( 6.493) | ( 2.045) | ( 0.435) |
| $h = -2$ | - | - | - | - |
| $h = -1$ | 0 | 0 | 0 | 0 |
| | | [Normalization horizon] | | |
| $h = 0$ | 0.128 | 9.121 | 0.464 | 0.274 |
| | ( 0.961) | ( 8.244) | ( 1.500) | ( 0.544) |
| $h = 1$ | 2.120 | 10.796 | 1.012 | 0.009 |
| | ( 2.189) | ( 8.456) | ( 1.552) | ( 1.105) |
| $h = 2$ | 1.388 | 8.216 | 3.487 | -0.879 |
| | ( 2.320) | (12.770) | ( 3.560) | ( 1.451) |
| $h = 3$ | 0.328 | 3.060 | 1.996 | -0.736 |
| | ( 3.008) | (17.957) | ( 3.884) | ( 1.010) |
| $h = 4$ | 0.238 | 0.539 | 2.198 | 0.326 |
| | ( 3.584) | (25.887) | ( 3.613) | ( 0.846) |
| $h = 5$ | 0.996 | 3.904 | 2.384 | 1.769 |
| | ( 4.887) | (33.913) | ( 4.289) | ( 1.374) |
| Median $N$ | 1240 | 1244 | 1238 | 902 |

*Notes:* LP-DiD estimates include country and year fixed effects; standard errors clustered at the country level in parentheses. The control group is restricted to SSA countries. $h = -1$ is the normalisation horizon and is zero by construction. Columns (1) and (2) report cumulative responses in log points × 100; columns (3) and (4) report the response of the ratio to GDP in percentage points at horizon $h$, which is a level gap rather than a cumulative sum. $^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$. Figures report 90% confidence intervals.

Table 10: Fiscal mechanisms: response to a refugee inflow shock (SSA Countries Only)

| | Expenditure (% of GDP) | Revenue (% of GDP) | Tax revenue (% of GDP) | Non-tax revenue (% of GDP) |
|---|---|---|---|---|
| Horizon | (1) | (2) | (3) | (4) |
| $h = -3$ | 1.138*** | -0.399 | -0.397 | 0.063 |
| | ( 0.413) | ( 0.347) | ( 0.336) | ( 0.305) |
| $h = -2$ | - | - | - | - |
| $h = -1$ | 0 | 0 | 0 | 0 |
| | | [Normalization horizon] | | |
| $h = 0$ | 0.796 | 1.026* | 0.513** | 0.391 |
| | ( 0.677) | ( 0.578) | ( 0.235) | ( 0.411) |
| $h = 1$ | 1.411 | 1.317* | 1.478*** | -0.366 |
| | ( 1.329) | ( 0.758) | ( 0.387) | ( 0.373) |
| $h = 2$ | 2.869* | 2.086*** | 1.945*** | -0.173 |
| | ( 1.557) | ( 0.782) | ( 0.662) | ( 0.575) |
| $h = 3$ | 2.224** | 1.816 | 1.793* | -0.368 |
| | ( 0.983) | ( 1.367) | ( 0.924) | ( 0.840) |
| $h = 4$ | 0.660 | 1.465 | 1.212 | -0.352 |
| | ( 1.110) | ( 1.591) | ( 0.956) | ( 1.056) |
| $h = 5$ | -0.655 | 1.536 | 1.745* | -0.387 |
| | ( 2.330) | ( 2.117) | ( 1.054) | ( 1.252) |
| Median $N$ | 918 | 906 | 881 | 876 |

*Notes:* LP-DiD estimates include country and year fixed effects; standard errors clustered at the country level in parentheses. The control group is restricted to SSA countries. $h = -1$ is the normalisation horizon and is zero by construction. All outcomes are expressed as a share of GDP; coefficients report the response in percentage points at horizon $h$. $^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$. Figures report 90% confidence intervals.

Table 11: LP-DiD estimates with conflict and disaster controls (Full Sample)

| Horizon | Real GDP (log pts × 100) (1) | Consumer prices (log pts × 100) (2) | Current account (% of GDP) (3) | Primary balance (% of GDP) (4) |
|---|---|---|---|---|
| $h = -3$ | -1.135 | -3.284 | 0.843 | -0.680* |
| | ( 1.401) | ( 6.108) | ( 1.969) | ( 0.394) |
| $h = -2$ | — | — | — | — |
| $h = -1$ | 0 | 0 | 0 | 0 |
| | | [Normalization horizon] | | |
| $h = 0$ | -1.021 | 10.562 | 0.219 | 0.106 |
| | ( 1.006) | (10.054) | ( 1.714) | ( 0.512) |
| $h = 1$ | 0.678 | 15.912 | 0.361 | 0.142 |
| | ( 2.551) | (14.856) | ( 1.914) | ( 1.050) |
| $h = 2$ | -1.194 | 17.323 | 3.341 | -0.593 |
| | ( 2.995) | (19.072) | ( 4.507) | ( 1.487) |
| $h = 3$ | -3.229 | 19.024 | 1.851 | 0.940 |
| | ( 3.827) | (23.757) | ( 4.939) | ( 0.881) |
| $h = 4$ | -3.791 | 17.171 | 3.120 | 1.536** |
| | ( 4.570) | (28.529) | ( 5.222) | ( 0.763) |
| $h = 5$ | -2.793 | 24.547 | 3.949 | 1.703 |
| | ( 6.182) | (39.092) | ( 6.175) | ( 1.186) |
| Median $N$ | 2853 | 2824 | 2828 | 2043 |

*Notes:* LP-DiD estimates with controls for host-country conflict intensity (UCDP/PRIO) and natural disaster exposure (EM-DAT), both entered with a one-year lag so as to be predetermined with respect to the shock. Country and year fixed effects; standard errors clustered at the country level in parentheses. $h = -1$ is the normalisation horizon and is zero by construction; the estimate at $h = -2$ is zero by construction because the dependent variable at that horizon is the negative of the lagged first difference included as a control. Columns (1) and (2) report cumulative responses in log points × 100; columns (3) and (4) report the response of the ratio to GDP in percentage points at horizon $h$, which is a level gap rather than a cumulative sum. $^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$. Figures report 90% confidence intervals.

Table 12: Fiscal mechanisms: with conflict and disaster controls (Full Sample)

| | Expenditure (% of GDP) | Revenue (% of GDP) | Tax revenue (% of GDP) | Non-tax revenue (% of GDP) |
|---|---|---|---|---|
| Horizon | (1) | (2) | (3) | (4) |
| $h = -3$ | 0.685** | -0.420 | -0.254 | 0.037 |
| | ( 0.302) | ( 0.376) | ( 0.267) | ( 0.218) |
| $h = -2$ | - | - | - | - |
| $h = -1$ | 0 | 0 | 0 | 0 |
| | | [Normalization horizon] | | |
| $h = 0$ | 1.113* | 1.406*** | 0.606** | 0.554 |
| | ( 0.588) | ( 0.519) | ( 0.235) | ( 0.370) |
| $h = 1$ | 1.325 | 1.590*** | 1.232*** | 0.104 |
| | ( 1.076) | ( 0.568) | ( 0.323) | ( 0.326) |
| $h = 2$ | 2.522* | 2.423*** | 1.603** | 0.444 |
| | ( 1.464) | ( 0.679) | ( 0.647) | ( 0.402) |
| $h = 3$ | 1.520 | 2.184* | 1.537* | 0.449 |
| | ( 0.947) | ( 1.275) | ( 0.903) | ( 0.599) |
| $h = 4$ | 0.394 | 1.996 | 1.076 | 0.619 |
| | ( 1.004) | ( 1.334) | ( 0.840) | ( 0.754) |
| $h = 5$ | 0.220 | 2.826* | 1.944** | 0.826 |
| | ( 1.763) | ( 1.596) | ( 0.908) | ( 0.937) |
| Median $N$ | 2184 | 2282 | 1996 | 1992 |

*Notes:* LP-DiD estimates with controls for host-country conflict intensity (UCDP/PRIO) and natural disaster exposure (EM-DAT), both entered with a one-year lag so as to be predetermined with respect to the shock. Country and year fixed effects; standard errors clustered at the country level in parentheses. $h = -1$ is the normalisation horizon and is zero by construction; the estimate at $h = -2$ is zero by construction because the dependent variable at that horizon is the negative of the lagged first difference included as a control. Results report the response of the ratio to GDP in percentage points at horizon $h$, which is a level gap rather than a cumulative sum. $^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$. Figures report 90% confidence intervals.

Table 13: Pooled five-year LP-DiD ATT on log real GDP under the total-displacement treatment definitions.

| Treatment definition | ATT | SE | $t$ |
|---|---|---|---|
| $T_2$: total displaced $\geq 5\%$ | $-0.120$ | 0.034 | $-3.55$ |
| $T_2'$: total displaced $\geq 7\%$ | $-0.243$ | 0.036 | $-6.71$ |

*LP-DiD with country and year fixed effects, outcome-specific lag controls, clustered standard errors at the country level. Pooled five-year ATT averages event-time horizons $h = 0$ through $h = 5$.*

Table 14: Summary Statistics of Key Variables for Burkina Faso Case Study

| Variable | Obs. | Mean | Std. Dev. | Min | Max |
|---|---|---|---|---|---|
| Annual Wage (CFAF) | 4,204 | 141,391.8 | 573,426.8 | 0 | 8,000,000 |
| Annual consumption per capita (CFAF) | 4,204 | 382,040.8 | 342,393.4 | 43,864.85 | 4,081,843 |
| Employment Status over the previous 12 months | | | | | |
| Employed | 4,204 | 0.627 | 0.484 | 0 | 1 |
| Household work | 4,204 | 0.100 | 0.300 | 0 | 1 |
| Non-Employed | 4,204 | 0.272 | 0.445 | 0 | 1 |
| Time | 4,204 | 0.519 | 0.500 | 0 | 1 |
| Treatment Group | 4,204 | 0.608 | 0.488 | 0 | 1 |
| Time × Treatment | 4,204 | 0.276 | 0.447 | 0 | 1 |
| Years of experience | 4,204 | 24.245 | 12.549 | 0 | 43 |
| Education level | | | | | |
| No education | 4,204 | 0.735 | 0.441 | 0 | 1 |
| Primary education completed | 4,204 | 0.155 | 0.362 | 0 | 1 |
| Secondary education completed | 4,204 | 0.094 | 0.292 | 0 | 1 |
| Post-secondary education completed | 4,204 | 0.016 | 0.124 | 0 | 1 |
| Proportion IDPs in the city | 4,204 | 2.090 | 5.354 | 0 | 23.830 |
| Gender (1=Male) | 4,204 | 0.450 | 0.500 | 0 | 1 |

Table 15: IDP Flows and Wages in Burkina Faso

| | Wages (log) | | | | | |
|---|---|---|---|---|---|---|
| | (1) (Overall) | (2) (Agriculture) | (3) (Industry) | (4) (Services) | (5) (Rural) | (6) (Urban) |
| Time | -2.025 | 1.434 | -16.265*** | 6.201*** | -2.687 | -1.520 |
| | (2.147) | (1.151) | (2.008) | (1.423) | (3.041) | (1.967) |
| Treatment Group | -2.237 | -1.241 | -9.284*** | -0.591 | -4.038 | 0.761 |
| | (2.095) | (0.795) | (1.895) | (1.368) | (2.969) | (1.904) |
| Treatment X Time | 1.790 | -0.632 | 14.218*** | -5.536*** | 2.451 | 0.887 |
| | (2.144) | (1.150) | (2.423) | (1.773) | (3.058) | (2.031) |
| Primary education | 1.530 | -3.032* | 1.971* | 3.021*** | 2.786 | 2.371*** |
| | (1.487) | (1.778) | (1.149) | (1.002) | (2.053) | (0.898) |
| Secondary education | 7.062*** | -1.429 | 10.507*** | 11.442*** | 7.969*** | 7.842*** |
| | (1.848) | (1.051) | (1.491) | (1.144) | (2.374) | (1.346) |
| Post secondary education | 9.482*** | 0.551 | | 12.050*** | | 8.846*** |
| | (1.765) | (0.811) | | (1.945) | | (1.560) |
| Years of experience | 0.596*** | 0.103 | 0.338* | 0.388** | 1.047*** | 0.267 |
| | (0.163) | (0.112) | (0.195) | (0.172) | (0.245) | (0.195) |
| Years of experience squared | -0.011*** | -0.003* | -0.008** | -0.007** | -0.019*** | -0.002 |
| | (0.003) | (0.002) | (0.004) | (0.003) | (0.005) | (0.005) |
| Regional dummies | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 2588 | 1215 | 257 | 443 | 1694 | 894 |
| R-squared | 0.283 | 0.338 | 0.740 | 0.726 | 0.368 | 0.326 |

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 16: IDP flows and labor force participation in Burkina Faso

| | **Probability of being employed** | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| | Overall | Rural | Urban |
| Time | 0.059 | -0.012 | 0.074** |
| | (0.031) | (0.053) | (0.034) |
| Treatment Group | 0.145*** | 0.097* | 0.082* |
| | (0.035) | (0.055) | (0.050) |
| Treatment X Time | -0.135*** | -0.033 | -0.173*** |
| | (0.042) | (0.075) | (0.046) |
| Primary education | 0.202*** | 0.224*** | 0.125*** |
| | (0.017) | (0.026) | (0.027) |
| Secondary education | 0.236*** | 0.268*** | 0.184*** |
| | (0.019) | (0.036) | (0.025) |
| Post secondary education | 0.337*** | | 0.309*** |
| | (0.024) | | (0.032) |
| Years of experience | 0.050*** | 0.038*** | 0.059*** |
| | (0.003) | (0.004) | (0.003) |
| Years of experience squared | -0.001*** | -0.001*** | -0.001*** |
| | (0.000) | (0.000) | (0.000) |
| Regional dummies | Yes | Yes | Yes |
| Observations | 4,204 | 2,597 | 1,607 |

Average marginal effects with standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 17: IDP flows and consumption per capita in Burkina Faso

| | Consumption per capita (Log) | | | | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| | Overall | Agriculture | Industry | Services | Rural | Urban |
| Time | 0.219* | 0.774*** | 0.264 | 0.265** | 0.324*** | 0.115 |
| | (0.113) | (0.220) | (0.309) | (0.113) | (0.104) | (0.151) |
| Treatment Group | -0.154 | 0.049 | 0.646** | 0.176 | -0.238*** | 0.009 |
| | (0.107) | (0.196) | (0.293) | (0.118) | (0.077) | (0.134) |
| Treatment X Time | 0.020 | -0.553** | -0.343 | -0.239* | -0.005 | -0.061 |
| | (0.110) | (0.221) | (0.342) | (0.141) | (0.111) | (0.169) |
| Primary education | 0.252* | 0.527*** | 0.362*** | 0.321*** | -0.018 | 0.299*** |
| | (0.144) | (0.165) | (0.114) | (0.077) | (0.090) | (0.081) |
| Secondary education | 0.762*** | 0.501** | 1.050*** | 0.627*** | 0.567*** | 0.452*** |
| | (0.105) | (0.200) | (0.113) | (0.116) | (0.153) | (0.083) |
| Post secondary education | 1.133*** | 0.778*** | 0.962*** | 0.818*** | | 0.881*** |
| | (0.144) | (0.103) | (0.257) | (0.144) | | (0.119) |
| Years of experience | 0.020 | 0.017* | 0.052*** | 0.029** | 0.014* | 0.008 |
| | (0.013) | (0.010) | (0.016) | (0.013) | (0.008) | (0.006) |
| Years of experience squared | 0.000** | -0.000* | -0.000 | -0.000* | -0.000*** | -0.000*** |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) |
| Regional dummies | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 2588 | 1215 | 257 | 443 | 1694 | 894 |
| R-squared | 0.471 | 0.671 | 0.763 | 0.569 | 0.649 | 0.406 |

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 18: IDP flows and inflation in Burkina Faso

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | Overall food index | Maize | Millet | Sorghum |
| Time | 0.034 | 0.148*** | 0.001 | -0.030 |
| | (0.026) | (0.025) | (0.033) | (0.032) |
| Treatment Group | 0.047* | 0.070*** | 0.037 | 0.039 |
| | (0.024) | (0.025) | (0.029) | (0.029) |
| Treatment X Time | -0.015 | -0.037 | -0.006 | -0.015 |
| | (0.033) | (0.033) | (0.040) | (0.041) |
| Regional dummies | Yes | Yes | Yes | Yes |
| Observations | 128 | 128 | 128 | 128 |
| R-squared | 0.538 | 0.708 | 0.429 | 0.493 |

Standard errors in parentheses; Observations are municipalities

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 19: South Sudan - Summary Statistics

| Variable | Obs. | Mean | Std. Dev. | Min | Max |
|---|---|---|---|---|---|
| Refugees | 1054 | 2704.48 | 8318.89 | 0 | 69722 |
| Returnees | 1054 | 13500.10 | 24017.44 | 0 | 135441 |
| County-level population (2023) | 1054 | 217681.97 | 188868.97 | 38976 | 807040 |
| Price index (log) | 1054 | 2.78 | 0.76 | 1.31 | 4.28 |
| Proportion of refugees and asylum-seekers | 1054 | 0.03 | 0.17 | 0.00 | 1.79 |
| Proportion of returnees | 1054 | 0.13 | 0.38 | 0.00 | 2.52 |
| Proportion of refugees, asylum-seekers, and returnees | 1054 | 0.15 | 0.52 | 0.00 | 4.30 |
| Treatment (refugees and asylum-seekers) | 1054 | 0.05 | 0.23 | 0.00 | 1.00 |
| Treatment (returnees) | 1054 | 0.31 | 0.46 | 0.00 | 1.00 |
| Treatment (refugees, asylum-seekers, and returnees) | 1054 | 0.34 | 0.47 | 0.00 | 1.00 |

Table 20: Summary Statistics of Key Variables for Uganda Case Study

| Variable | Obs. | Mean | Std. Dev. | Min | Max |
|---|---|---|---|---|---|
| Wage (CFAF) | 22,719 | 839,665.7 | 9,893,948 | 1 | $1.38\times10^9$ |
| Consumption per capita (UGX) | 34,426 | 95,685.86 | 972,409.2 | 3,380.60 | $1.26\times10^8$ |
| Employed (last month) | 22,684 | 0.686 | 0.464 | 0 | 1 |
| Time | 47,182 | 0.679 | 0.467 | 0 | 1 |
| Treatment group | 47,182 | 0.128 | 0.334 | 0 | 1 |
| Time × Treatment | 47,182 | 0.089 | 0.284 | 0 | 1 |
| Years of experience | 47,182 | 27.08 | 12.95 | 0 | 43 |
| *Education level* | | | | | |
| No education | 34,994 | 0.192 | 0.394 | 0 | 1 |
| Primary education | 34,994 | 0.490 | 0.500 | 0 | 1 |
| Secondary education | 34,994 | 0.248 | 0.432 | 0 | 1 |
| Post-secondary education | 34,994 | 0.071 | 0.256 | 0 | 1 |
| Share of Refugees in districts (%) | 47,182 | 2.77 | 16.48 | 0 | 34.65 |
| Male (=1) | 47,182 | 0.482 | 0.500 | 0 | 1 |

Table 21: Refugees flows and wages in Uganda

| | Wages (Log) | | | | | |
|---|---|---|---|---|---|---|
| | (1) Overall | (2) Agriculture | (3) Industry | (4) Services | (5) Urban | (6) Rural |
| Time | -0.102 | -1.029*** | 0.381 | 0.703** | 0.554** | -0.323*** |
| | (0.116) | (0.130) | (0.574) | (0.299) | (0.280) | (0.125) |
| Treatment group | 0.680*** | 0.475** | 0.109 | 0.812** | 0.638*** | 0.555*** |
| | (0.155) | (0.238) | (0.528) | (0.339) | (0.240) | (0.208) |
| Treatment × Time | 0.098 | -0.132 | 0.207 | -0.260 | 0.175 | 0.174 |
| | (0.203) | (0.278) | (0.689) | (0.462) | (0.330) | (0.259) |
| Primary education | -0.350*** | -0.568*** | 0.069 | -1.518*** | -0.586** | -0.244*** |
| | (0.079) | (0.095) | (0.540) | (0.353) | (0.241) | (0.082) |
| Secondary education | 0.459*** | -1.119*** | 0.696 | -0.089 | -0.462* | 0.741*** |
| | (0.105) | (0.121) | (0.557) | (0.373) | (0.259) | (0.116) |
| Post secondary education | 4.405*** | 0.132 | 2.525*** | 3.899*** | 2.754*** | 5.578*** |
| | (0.181) | (0.325) | (0.591) | (0.412) | (0.321) | (0.248) |
| Years of experience | 0.139*** | 0.072*** | 0.133*** | -0.157*** | 0.205*** | 0.050 |
| | (0.012) | (0.014) | (0.049) | (0.035) | (0.065) | (0.074) |
| Years of experience squared | -0.003*** | -0.002*** | -0.005*** | 0.001 | -0.005*** | -0.003*** |
| | (0.000) | (0.000) | (0.001) | (0.001) | (0.001) | (0.000) |
| Regional and area dummies | Yes | Yes | Yes | Yes | Yes | Yes |
| Time fixed effects | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 34768.000 | 18343.000 | 2493.000 | 5814.000 | 8519.000 | 26249.000 |
| R-squared | 0.121 | 0.048 | 0.187 | 0.145 | 0.117 | 0.095 |

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 22: Refugee flows and labor force participation in Uganda

| | **Probability of being employed** | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| | Overall | Rural | Urban |
| Time | -0.307*** | -0.188*** | -0.335*** |
| | (0.005) | (0.011) | (0.005) |
| Treatment Group | -0.152*** | -0.134*** | 0.012 |
| | (0.010) | (0.015) | (0.024) |
| Treatment × Time | 0.170*** | 0.145*** | -0.005 |
| | (0.012) | (0.019) | (0.025) |
| Primary education | -0.147*** | -0.028 | -0.183*** |
| | (0.006) | (0.018) | (0.009) |
| Secondary education | -0.105*** | 0.004 | -0.148*** |
| | (0.007) | (0.020) | (0.011) |
| Post-secondary education | -0.033*** | 0.079*** | -0.057*** |
| | (0.009) | (0.022) | (0.016) |
| Years of experience | 0.016*** | 0.023*** | 0.013*** |
| | (0.001) | (0.001) | (0.001) |
| Years of experience squared | -0.00031*** | -0.00042*** | -0.00025*** |
| | (0.00002) | (0.00003) | (0.00002) |
| Regional dummies | Yes | Yes | Yes |
| Observations | 34,768 | 26,249 | 8,519 |

Average marginal effects with standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 23: Refugees flows and consumption per capita in Uganda

| | Consumption per capita (Log) | | | | | |
|---|---|---|---|---|---|---|
| | (1) Overall | (2) Agriculture | (3) Industry | (4) Services | (5) Urban | (6) Rural |
| Time | 0.473*** | 0.518*** | 0.441*** | 0.412*** | 0.469*** | 0.482*** |
| | (0.008) | (0.010) | (0.029) | (0.021) | (0.008) | (0.018) |
| Treatment Group | 0.272*** | 0.163*** | 0.215*** | 0.267*** | 0.122*** | 0.336*** |
| | (0.016) | (0.028) | (0.049) | (0.034) | (0.022) | (0.025) |
| Treatment × Time | -0.137*** | -0.173*** | -0.056 | -0.112** | -0.095*** | -0.118*** |
| | (0.021) | (0.034) | (0.066) | (0.045) | (0.027) | (0.034) |
| Primary | 0.259*** | 0.252*** | 0.134** | 0.157*** | 0.277*** | 0.168*** |
| | (0.010) | (0.013) | (0.058) | (0.042) | (0.011) | (0.031) |
| Secondary | 0.581*** | 0.506*** | 0.480*** | 0.489*** | 0.568*** | 0.558*** |
| | (0.013) | (0.017) | (0.060) | (0.043) | (0.014) | (0.033) |
| Post secondary education | 1.106*** | 1.006*** | 0.925*** | 0.930*** | 1.101*** | 1.054*** |
| | (0.019) | (0.042) | (0.066) | (0.048) | (0.027) | (0.037) |
| Years of experience | 0.009*** | 0.003 | -0.004 | 0.016*** | 0.004** | 0.019*** |
| | (0.001) | (0.002) | (0.005) | (0.004) | (0.002) | (0.003) |
| Years of experience squared | -0.000** | 0.000* | 0.000 | -0.000*** | 0.000 | -0.000*** |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) |
| Regional and district dummies | Yes | Yes | Yes | Yes | Yes | Yes |
| Time fixed effects | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 34,734 | 18,326 | 2,491 | 5,811 | 26,220 | 8,514 |
| R-squared | 0.376 | 0.278 | 0.375 | 0.321 | 0.268 | 0.322 |

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 24: Refugee flows and inflation in Uganda

| | Unit prices (Log) | | | |
|---|---|---|---|---|
| | (1) Rice | (2) Corn | (3) Sugar | (4) Tea |
| Time | 0.502*** | 0.626*** | 0.425*** | 2.557*** |
| | (0.017) | (0.018) | (0.010) | (0.086) |
| Treatment group | 0.184*** | 0.293*** | -0.041 | 0.105 |
| | (0.051) | (0.085) | (0.033) | (0.081) |
| Treatment × Time | -0.047* | -0.070** | 0.142*** | 0.981*** |
| | (0.026) | (0.030) | (0.019) | (0.108) |
| Regional and district dummies | Yes | Yes | Yes | Yes |
| Time fixed effects | Yes | Yes | Yes | Yes |
| Observations | 30,592 | 40,100 | 57,879 | 8,685 |
| R-squared | 0.185 | 0.170 | 0.195 | 0.389 |

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$



PUBLICATIONS

**Macroeconomic Effects of Forced Displacement in Sub-Saharan Africa**
Working Paper No. [WP/2026/228